SECTION 14210 - HYDRAULIC ELEVATORS

PART 1 GENERAL

Related Sections include the following:

Division 2 Section "Earthwork" for excavation to accommodate plunger-cylinder assembly.
Division 3 Section "Cast-in-Place Concrete" for setting sleeves, inserts, and anchoring devices in concrete.
Division 4 Section "Unit Masonry" for setting sleeves, inserts, and anchoring devices in masonry.
Division 5 Section "Structural Steel" for the following:
Divider beams.
Division 5 Section "Metal Fabrications" for the following:
Attachment plates and angle brackets for supporting guide-rail brackets.
Structural-steel shapes for subsills.
Pit ladders.
Division 9 Section "Resilient Tile Flooring" for finish flooring in elevator cars.
Division 13 Section "Fire Alarm" for smoke detectors in elevator lobbies to initiate emergency recall operation and for connection to elevator controllers.
Division 16 Section "Premises Telephone Wiring" for telephone service to elevators.
Division 16 Sections for electrical service for elevators to and including fused disconnect switches at machine room door.

1.1 SUMMARY:

One New Hydraulic Passenger Elevators: This specification covers the complete furnishing and installing of equipment required to install one new hydraulic passenger elevator at Southeastern Louisiana University, Kinesiology & Health Studies Addition.

Elevator Contractor shall furnish all labor, tools, materials, fixtures, equipment, accessories, doors and door frames, elevator cabs, transportation, controllers, motors, hoistway wiring, power unit, hydraulic piping system, jack and plunger, casing, including all electrical and mechanical systems including necessary auxiliaries, etc., to complete the hydraulic passenger elevator installation as required by this specification that meets or exceeds the requirements of ASME A17.1, NFPA 70, and with the ADAAG (Americans with Disabilities Act Architectural Guidelines).

1.2 DEFINITIONS:

A. HYDRAULIC ELEVATORS are hereby defined to include systems in which cars are hoisted either directly or indirectly by action of a hydraulic plunger and cylinder (jack); car enclosures, car and hoistway doors with related accessories, hoistway entrances, operating, dispatching, safety, security, leveling, alarm, maintenance, and similar required performances and capabilities.
B. CODE when referred to in these specifications is the ASME A17.1 – 2000 with addenda, Safety Code for Elevators and Escalators.
C. CONTRACTOR is defined as the Elevator Contractor.
D. ENGINEER is defined as the Project Engineer or Project Architect.
E. PROVIDE as used in these specifications and on the Drawing shall be termed to mean " furnish and install".
F. OWNER when referred to in these specifications shall mean Southeastern Louisiana University.
G. STATE when referred to in these specifications shall mean "The State of Louisiana".
H. ADA or ADAAG is herein defined as the Americans with Disabilities Act Architectural Guidelines.
I. SINGULAR NUMBER: In cases where a device or part of the equipment is herein referred to in the singular number, it is intended that such reference shall apply to as many such devices as are required to complete installation.

1.3 SYSTEM PERFORMANCE REQUIREMENTS:

A. The elevators shall be capable of meeting the highest standards of the industry. Elevator schedules indicate required performances, controls, capacities, features, and finishes for each elevator or group of elevators and are included in these specifications.

B. The pump motor shall provide continuous service of not less that 80 starts per hour, and with this service, the motor windings shall not exceed 55 degrees C. ambient temperature rise. Speed regulation, when the car is carrying its rated load, shall not exceed plus or minus 5% of the average trip speed.

C. Starting, stopping and leveling shall be smooth and comfortable without appreciable steps of acceleration or deceleration. Stopping shall be without bumps or jars.

D. Cars shall not move side to side during process of opening and closing doors.

1.4 INTENDED ENVIRONMENTAL CONDITIONS: For informational purposes, the environmental conditions expected to occur within the Elevator Equipment Room and the Elevator Shaft shall be as indicated herein. Assure that the equipment furnished will operate satisfactorily within the prescribed conditions. If any modifications are required, it will be the responsibility of the Elevator Contractor to make any necessary changes. If any equipment is furnished that will not properly function within established parameters, pay any additional costs attributable to providing proper equipment.

A. ELEVATOR EQUIPMENT ROOMS: The mechanical systems for these Areas are intended to have 50 degree F. Air leaving the registers. The maximum temperature that should be considered for the rooms should be considered at 100 degrees F. Relative humidity is intended to be 60% maximum.

B. HEAT OUTPUT OF ELEVATOR EQUIPMENT: Upon selection of the elevator equipment, furnish the maximum BTU/HR output of the proposed equipment to the ENGINEER for possible adjustment of the entering air temperature into the rooms, if necessary.

1.5 REFERENCES: The publications listed below form a part of the specifications to the extent referenced. The publications are referred to in the text by basic designation only. All work shall be performed in accordance with the latest revised edition of the below referenced codes (as of the date the bids are taken)

AIR CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

APPLICABLE FEDERAL STATE AND MUNICIPAL CODE AND REQUIREMENTS

AMERICAN DISABILITY ACT (ADA) 1992 EDITION

AMERICAN STANDARDS INSTITUTE (ANSI)

ANSI A117.1 Providing Accessibility and Usability for Physically Handicapped People

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 366 Steel, Sheet, Carbon, Cold-Rolled, Commercial Quality ASTM E 152 Fire Tests of Door Assemblies

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A 17.1 (2000 and addenda) Safety Code for Elevator and Escalators
ASME A17.2 (2001) Inspector's Manual for Hydraulic Elevator

ELECTRICAL TESTING LABORATORIES (ETL)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA (IES)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA LD 3 High Pressure Decorative Laminates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (1998) National Electrical Code

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA)

SHEET METAL & AIR CONDITIONING CONTRACTORS= NATIONAL ASSOCIATION (SMACNA)

STANDARD BUILDING CODE

UNDERWRITER-S LABORATORIES (UL)

1.6 **DESCRIPTION AND SCOPE OF WORK:** The work under this section is subject to those provisions of the General and Special Conditions, which in any way affect the work herein specified.

This specification guideline is intended to cover the installation of one new hydraulic passenger elevator at the Southeastern Louisiana University - Kinesiology & Health Studies Addition.

Perform all work under this heading, including the furnishing of labor, materials, equipment, and service necessary for and reasonably incidental to the proper completion of all mechanical and electrical work as shown on the drawings and herein specified.

Provide manufacturer's standard per-engineered elevator systems that will comply with or fulfill the requirements of elevator schedule sheets in these specifications. The manufacturers option, provide custom-manufactured elevator systems that will fulfill requirements. Where components are not otherwise indicated, provided standard components published by manufacturer as included in standard pre-engineered elevator systems and as required for a complete system.

The CONTRACTOR shall submit a Progress/Phasing Plan to the Project Administrator for approval. The Project Administrator will schedule a final inspection with the Elevator Inspection Agency (ETS) on Contract with Southeastern Louisiana University within this time frame.

A mandatory Site Visit is required of all Bidders on this Project. Bidders may examine the job site and confer with all authorities concerned in order to become familiar with all existing conditions pertinent to the work to be performed. No additional compensation will be allowed for failure to be so informed.

Materials and equipment shall be new, of the best quality, with same brand or manufacturer for all similar material.

All work shall be performed in a neat and workmanlike manner and is to include all work and materials in accordance with the drawings and as specified ,etc.

In all cases where a device or part of the equipment is herein referred to in the singular number, it is intended that such reference shall apply to as mach such devices as are required to complete the installation.

1.7 **Specification Explanation:** These specifications are of the abbreviated, simplified or streamline type and include incomplete sentences. Omission of words or phrases such as "The Contractor shall", "in conformity therewith", "shall be", "as noted on the drawings", "according to the plans", "a", "the", "all", etc., are intentional. Omitted words or phrases shall be supplied by inference in the same manner as they are when a note occurs on the drawings.

Regardless of titles and subdivisions herein employed, consider these specifications as one complete document

with General Sections applying to all other sections. All bidders are cautioned to read the entire specifications and to thoroughly familiarize themselves with all requirements thereto.

Check all specifications and/or drawings and bring to attention any conflicts or variations as soon as noted. Protect the State of Louisiana and his agents including the ENGINEER and elevator consultant from any and all damages and expense arising from fulfillment of contact and at completion of work, repair all damages done.

For any points which are not clear, or for items and/or details which the Elevator Contractor feels are in need of clarification, consult the ENGINEER before submission of a proposal.

In case of discrepancies and/or ambiguities in the drawings and/or in the specifications, consult ENGINEER prior to submission of a proposal. Failure to do so one the part of the successful bidder shall be construed as explicit agreement on his part to abide by the ENGINEER'S decision in such matters.

1.8 Quality Assurance:

Installer Qualifications: Engage the elevator manufacturer or elevator supplier who has completed elevator installations similar in material, design with a minimum of ten (10) years documented experience. The ELEVATOR CONTRACTOR bidding on this Project must be fully experienced in installations of equal size, complexity, and quality, and must be licensed to perform such work as required by the Louisiana State Legislature, R.S. 37:2152- 2163.

The Elevator Contractor bidding on this portion of the work must be fully experienced in installations of equal size, complexity, and quality, and must be licensed to perform such work by the Louisiana State Legislature, R.S. 37: 2152-2163.

In bidding he acknowledges that he fully understands the scope of the work and design, and has the ability for the contract price to assemble and install the equipment, piping, duct work, etc., shown or specified, so as to mold same into a satisfactory workable system and arrangement.

Contractor shall recognize that a fault or error in his work remains his responsibility regardless of whether such difficulty was discovered after the work had progressed, and shall make corrections at no cost to Southeastern Louisiana University.

1.9 Bidder's Qualifications

In the interest of unified responsibility, the ELEVATOR CONTRACTOR shall be one regularly engaged in the business of manufacturing, installing and servicing elevators of the type and character required by these specifications and he shall manufacture all components (signal fixtures and controller) and all other parts of the equipment as required by these specifications or shall be an approved installer of the manufacturer's equipment and he shall so state in his request for approval, listing the items to be supplied by the manufacturer. The successful ELEVATOR CONTRACTOR must be registered to do business in the State of Louisiana and its primary business is installing, maintaining and selling elevators of the type requested in these specification.

The Elevator Contractor must maintain a service representative within the local area insure a maximum of (02) two hour response time for call - backs.

All work shall be performed in accordance with the latest revised edition (as of the date bids are taken) of the Safety Code for Elevators and Escalators (ASME A 17.1), the National Electrical Code (NFPA 70), and /or such State and local elevator codes as may be applicable.

Recognize that amount if information and detail that could be provided in contact documents is limitless and could extend into every minute detail and sequence of operations, to a point where only workmen would be required, without drawing on ability, experience and ingenuity of the Elevator Contractor. Install all work in strict accordance, with all local and state codes and ordinances, with National Board of Fire Underwriters, and in accordance with all

OSHA Standards.

Secure all permits and inspections and pay all legal fees and taxes and provide The State of Louisiana with a certificate of approval from agencies having jurisdiction over various phases of the work.

Additionally the project is to adhere to the current requirements of the State of Louisiana Office of State Buildings and the Elevator Inspection Contractor.

1.10 Qualifications of Elevator Mechanics

Qualifications certificates of experience of Elevator mechanics employed to install supervise, test and maintain the elevator(s) shall certify mechanics to have not less than five (05) years of experience installing, supervising and testing elevators of the type and rating specified. Certificates of experience shall be furnished to the ENGINEER prior to installation of the elevator(s). Helpers or apprentices with less than five (05) years experience will be permitted to work under the direct supervision of an elevator mechanic.

1.11 JOB CONDITIONS:

1.11.1 Job conditions include any accompanying drawings, plans, diagrams, notes, etc., are shown to limit and explain structural conditions, construction requirements, sizes, capacities, and method of installation and erection, Structural and other conditions may require certain modifications and adjustments from conditions shown. Such deviations are permissible; however specific sizes, capacities and requirements affecting the satisfactory performance and operation of the installation shall remain unchanged. Make allowance for normal job conditions and interference's.

Whenever it becomes necessary to shift ducts or pipes or to change shapes of ducts such changes shall be referred to the ENGINEER for approval.

Ask for details whenever uncertain about method of installation. Lack of details not requested shall not excuse improper installation and correction shall be the responsibility of the Elevator Contractor.

Schedule and perform all mechanical work to avoid delays.

All piping, clean outs and covers, and other mechanical items in ways of construction, shall be re-routed, relocated or otherwise adjusted to work out with such construction or changes shown or specified in any or all various sections of specifications.

Refer unknown site of job conditions that are encountered immediately to the ENGINEER for method of disposition before continuation of work.

1.11.2 Verifications of Dimensions: Elevator Contractor shall become familiar with all details of the work, verify all dimensions in the field and advise the ENGINEER of any discrepancy before performing any work.

1.11.3 Use of Asbestos Products: Products which contain asbestos are prohibited. This prohibition includes items such as packing, gaskets, or linings for car guide shoes even though the item is encapsulated or the asbestos fibers are impregnated with binder material.

1.12 Fire & Safety:

A. Exits for this Modernization Project having occupied areas of the building including rooms, suites, corridors and floors shall not be blocked by the construction or by construction materials. Exits may be blocked temporarily if it is unavoidable and adequate alternate measures are provided such as signage, instructions to occupants and a heat detection system and with approval from the PROJECT ADMINISTRATOR.

B. Existing fire protection systems including fire alarms systems, smoke detection systems, and sprinkler systems shall not be impaired during construction except for devices in the immediate construction area.

C. At the end of each workday, combustible packaging and crating materials for building products and equipment to be installed shall be removed from the building.

D. The necessary number and appropriate type of portable fire extinguisher per National Fire Protection Association (NFPA) Standard Nos. 10 and 241 shall be provided for all construction areas.

E. All flammable liquids shall be handled, stored and used in accordance with NFPA Standard No. 30.

F. All temporary electrical wiring and equipment used for construction shall be installed and used in accordance with pertinent provisions of NFPA Standard No. 70.

G. Installation of sprinkler systems, standpipe systems, fire hydrants and fire alarm systems, shall be given priority and placed into service as soon as practical.

H. Maintain construction site to permit access of fire department vehicles as necessary. Clear building construction areas of unnecessary obstructions so that all portions are accessible for fire department apparatus and permit emergency egress of construction and other personnel.

I. All precautions necessary shall be taken by the contractor to prevent accidental operation of any existing smoke detectors by minimizing the amount of dust generated in the vicinity of any smoke detectors.

J. All construction activities not already covered above shall be in accordance with the latest edition of NFPA No. 241, Standard for Safeguarding Construction, Alteration, and Demolition Operations, in effect at time of contract award.

K. At no time will protected openings in vertical shafts be left unprotected without equivalent fire - rated construction. Openings which are a part of the construction project shall not be left unprotected when the Elevator Contractor is not actively working in the area or after normal working hours.

L. STORM PROTECTION: Should warnings of wind of gale force or stronger be issued, the Elevator Contractor shall take every practicable precaution to minimize danger to persons, to the work, and to adjacent property. These precautions shall include closing all openings, removing all loose materials, tools and equipment from exposed locations, and removing or securing scaffolding and other temporary work.

1.13 Operations and Storage Area for the Elevator Contractor:

A. The Elevator Contractor shall provide a Storage and Work Station Module (Construction Trailer). The Storage Module may be placed on a location in the parking lot to be assigned and approved by Southeastern Louisiana University Project Administrator. The Elevator Contractor shall supply the Storage Module without any additional cost to Southeastern Louisiana University for storage of his material and tools.

B. Workmen are subject to the rules of Southeastern Louisiana University applicable to their conduct.

C. Execute work so as to interfere as little as possible with normal functioning of Southeastern Louisiana University as a whole, including operations of utility services, fire protection systems and any existing equipment, and with work being done by others.

D. The Elevator Contractor shall take all measures and provide all material necessary for the protection of existing equipment and property in affected areas of construction against dust and debris, so that equipment and affected areas to be used by " Southeastern Louisiana University " operations will not be hindered.

E. In corridor areas, CONTRACTOR shall not store any materials and shall remove dust and debris continuously so as not to create any hazard or track dust into other areas.

F. No utility service such as water, gas, steam, sewers, electricity, fire protection systems and communication systems may be interrupted without the prior approval of the PROJECT ADMINISTRATOR.

G. Protection of the interior of existing structures at all times, from damage, dust and weather inclemency. Wherever work is performed, floor surfaces that are to remain in place shall be adequately protected prior to starting work, and this protection shall be maintained intact until all work in the area is completed.

H. If hazardous materials are encountered during demolition operations, comply with applicable regulations, laws, and ordinances concerning removal, handling, and protection against exposure of environmental pollution. Notify the PROJECT ADMINISTRATOR immediately when hazardous materials are encountered.

1.14 **Warranty, Inspection and Maintenance Service:** Warranty service shall be provided for the elevator for a period of 12 months after date of final acceptance by the User (Southeastern's Physical Plant). The elevator contractor shall furnish preventive maintenance, periodic testing, replacement parts and call back service in accordance with the State of Louisiana Elevator Maintenance Specifications and special conditions as specified at no cost and expense to Southeastern Louisiana University for 12 months.
In addition to any warranties that may be available under applicable law (and notwithstanding anything to the contrary contained elsewhere in this Contract or in any exhibit attached hereto), Contractor warrants to Owner that

A. all of the Contractor's Work shall be completed in accordance with these Specifications,
B. any and all machinery, equipment, controls, devices, materials, products and other items incorporated in or otherwise made a part of the Contractor's Work and/or delivered to Owner as part of Contractor's Work will be safe, meet all applicable safety standards, and will be free of any defects in design, construction, composition, workmanship, and any other defects, and
C. following Final Completion of all of the Contractor's Work, the Elevators will perform and continue to perform in accordance with the Specifications, ordinary wear and tear excepted.
D. Warranty and maintenance service shall be performed only by trained elevator mechanics and shall include manufacturer's warranty requirements including but not limited to adjusting, lubricating, repairing and cleaning of equipment and furnishing supplies and parts to keep elevators in operation, except parts made necessary by misuse, accident, or negligence not caused by Elevator Contractor.
E. Testing and adjustments shall be in accordance with applicable provisions of ASME A 17.1 and ASME A 17.2. Emergency call back service shall be included and available 8 hours a day, 5 days per week, (normal work week, with response time of one hour or less for a maintenance mechanic to be on site.
F. Elevator Contractor shall provide a maintenance mechanic to be available for all Inspections and Tests. Including tests for fire service, shunt trip, smoke - heat detectors and emergency power for each unit.

1.15 **SUBMITTALS:** Submit to ENGINEER for review, immediately after awards of contract, four (4) copies of complete descriptive information and dimensional data on all items of equipment, materials and accessories. Submit all shop drawings at one time. Piece submissions shall be acceptable.

1.15.2 DATA: Equipment and Performance Data including adequate descriptive literature and shop drawings shall be submitted for the following items consisting of load ratings, sizes, car and platform dimensions, design performance, operating characteristics, finishes, features of control system, speeds, car door opening and closing speeds, and similar information.

Hoistway and car doors
Operation and Control systems
Operating systems
Emergency operating and signaling
Terminal stopping devices
Electrical protective devices
Drive machine and pump unit
Electrical protective devices
Operating Devices

1.15.3 MANUFACTURER'S CATALOG DATA: Data shall be submitted for the following items:

Integrated Microprocessor Units
Group Controller Units
Hoistway and car doors
Hoistway entrances
Entrance frames
Entrance sills
Operation and control systems
Emergency operating devices
Telephones
Electrical protective devices
Electrical equipment
Paint materials
Cabs and interior design
Operating systems

1.15.4 DRAWINGS: The Elevator Contractor shall prepare drawings showing the general arrangements of the elevator equipment. These drawings shall be reviewed before proceeding with fabrication and installation of elevator. The connection diagrams and wiring diagrams shall be submitted for the Hydraulic Passenger Elevator System indicating the relations and connections of devices and apparatus by showing the general physical

layout of all controls, the interconnection of one system (or portion of system) with another, and internal tubing, wiring, and other devices.

Wiring diagram-detailing wiring for power, signal and control systems, differentiating clearly between manufacturer - installed wiring and field - installed wiring. Indicate maximum and average power demands.

SHOP DRAWINGS and submittal data shall be considered to be instruments of service only and submitted for the sole purpose of convenience to the CONTRACTOR to assist him in the performance of the contract. The ENGINEER'S review of the shop drawings and submittal data shall not supersede these specifications, the accompanying drawings or the contract terms, unless specifically and explicitly stipulated therein. Shop drawings shall include dimensional drawings showing plans, elevations, sections and large - scale details of car enclosures and hoistway entrances. Include elevator diagrams to indicate elevator services to each level and include excavation requirements for the jack.

It is the Elevator Contractor's responsibility to determine and verify field measurements, materials, field construction criteria, review each submittal and; check and coordinate each submittal with requirements of the work and Contract documents. The following shall be submitted in sufficient detail to show full compliance with the specifications:

Shop Drawings required by this paragraph of the specifications shall be submitted as a complete package in quadruplicate (4 prints) and separately and apart from materials specified under other Sections, and shall be marked as "REQUIRED SUBMITTAL". All submitted drawings and related elevator material should be forwarded to Southeastern Louisiana UniversityProject Engineer.

A. Complete dimensional layout in plan and elevation of the machine room, hoistway and pit showing the arrangements of the equipment and all pertinent details of each elevator unit specified including:
   1. Complete set of straight line wiring diagram showing all circuits nomenclature, etc., for the elevator.
B. Complete drawings of HOISTWAY doors and hangers showing details of construction and method of fastening to the entrances.
C. Dimensional drawings showing details of the car guide shoes and reactions at points of support and buffer impact loads.
D. Complete drawings of elevator car enclosure, showing dimensional details if construction, location of car equipment, fastenings to platform, car lighting, ventilation, ceiling framing and top exit.
E. Dimensional drawings showing details of all signal and car equipment and fixtures.
F. Complete drawings and characteristic curves of elevator pump and motor.
G. Dimensional drawings and description of power door operator, electronic infrared door detectors, etc.
H. Dimensional drawings and description of hoistway door hangers.
I. Cuts of drawings of hydraulic fluid scavenger pump system.
J. Drawings showing details of controllers, selectors, and supervisory panels and a complete dimensional drawing for the hydraulic fluid storage tank.
K. Location of circuit breaker disconnects switched, light switches and feeder connection points in the machine room.
L. Location of hoistway junction boxes for connection of car lights and telephone.
M. Cuts of drawings of Braille plates for car and hall push buttons and tactile jamb markings. Submit method of attachment.

1.15.5 WIRING DIAGRAMS:

A. Provide 3 complete sets of field wiring and straight-line wiring diagrams showings all electronic circuits of all elevator equipment, in the hoistway, as well as the machine room. The diagrams shall be supplied on 8-1/2" x 11"sheets and shall be prepared using elevator industry standard wiring diagram symbols and drafting techniques. They shall be framed on Plexiglas or on pivoted hardboard coated with an approved plastic sealer and mounted in elevator machine room as directed by ENGINEER. In the event modifications are found necessary during installation, diagrams shall be revised to include all corrections

made prior to and during the final inspection. Diagrams shall be delivered to the ENGINEER within 30 days of final acceptance ON DISK IN THE AUTO-CAD FORMAT.

B. The following information relating to the specific type of microprocessor controls in provided on this project shall be provided:

1. Owner information manual, containing general data on major components; maintenance and adjustment.
2. Base software listing
3. System logic description.
4. Device program tapes.
5. Operational Manual
6. Sequential Operations Manual
7. Input/Output Sheets
8. Password or Code for Access to Control Equipment
9. Complete wiring diagrams needed for field trouble shooting, adjustment, fault code interpretation, repair and/or replacement of components. Diagrams shall be base diagrams, containing all changes and additions made to the equipment during the design and construction period.
10. Changes made during the warranty period shall be noted on the drawings in adequate time to have the finalized drawings reproduced for mounting in the machine room no later than six months prior to expiration of the warranty.

1.15.6 SCHEMATICS: CONTRACTOR shall provide schematics of all major assemblies, sub-assemblies and modules. If such schematics are considered to be proprietary, the contractor shall not be allowed to bid on this project.

1.15.7 **SAMPLES AND DESCRIPTIVE DATA**: In accordance with provision of these guideline specifications, furnish the following: Materials shall be submitted separately and apart from materials specified under other Sections and shall be marked "REQUIRED SUBMITTAL".

A. Before executing any work, furnish information sufficient to evidence full compliance with contract requirements on proposed items. Such information shall include, as required: Manufacturer's Name, Trade Names, Model or Catalog Number, Nameplate Data (size, capacity, rating), and specification reference numbers.

B. Samples:

1. Four each stainless, 3-inches by 5-inches.
2. Four each plastic laminate, 3-inches by 5-inches.
3. Four each of color rubber floor tile, 12-inched square.
4. Four each of protection pads, 3-inches by 5-inches.
5. Four each of samples for exposed finishes of cars, hoist-way doors and frames refinishing work, and signal equipment: 3-inch-square samples of sheet materials; and 4-inch lengths of running trim members
6. No other samples of materials specified shall be submitted unless specifically requested after submission of manufacturer's name. In case such samples are furnished pursuant to request, adjustment in contract price and time will be made as provided.

C. Materials Data: Submit for approval on the shop drawings the name of manufacturer, type or style designation of the following items:

1. Controllers
2. Hydraulic power unit
3. HP and R.P.M. of hydraulic power unit motor
4. Electric control valves, including capacity range.
5. Electric power door operator; HP rating & R.P.M. of motor.
6. Hoistway and car doors
7. Hoistway door interlocks and electric contracts.
8. Stroke, certified maximum and minimum loads, and maximum striking speed of car buffers.
9. HP and CFM rating on cab ventilation unit
10. Scavenger line and pump
11. Cab emergency lighting fixture and rechargeable battery unit.
12. Infrared electronic door protection devices.

13. Cab lighting: wattage and ballast rating
14. Operating and control system
15. Integrated Microprocessor Units
16. Operating devices
17. Electrical protective devices
18. Electrical equipment
19. Cab and interior design
20. Paint and materials.
21. Jack Assembly

D. Certificates: Certificates of Compliance shall be submitted certifying that each product used complies with ASME A 17.1. Rule 1101, and has been tested to those requirements by an independent testing laboratory within a two-year period.

E. MANUALS: After completion of project, turn over to the Project Administrator three (3) bound sets of complete indexed operating and maintenance instructions. The manuals shall give complete instructions for operation, inspection, testing, adjusting, repair and maintenance of each system, including equipment malfunction check lists and code fault interpretations and floor setup tables. .Documentation supplied with each item shall include general owner's manuals, detailed service manuals and all schematics of all major assemblies, sub-assemblies and modules. If such schematics are considered to be proprietary, then the Contractor shall not be allowed to bid on this project.

1.16 SUBSTITUTIONS: All material, equipment, methods and accessories entering into the work under this section of contract are subject to approval or disapproval of the PROJECT ADMINISTRATOR. Approval of any manufacturer, material, or product, shall not constitute a waiver of PROJECT ADMINISTRATOR'S right to demand full compliance with contract requirements, including shape, size, quality and performance.

Equality of materials is that established by opinion of the PROJECT ADMINISTRATOR. Decision of the PROJECT ADMINISTRATOR is final. Acceptable elevator installation companies that meet these requirements are as follows:

Kone Elevator Company
Otis Elevator Company
Schindler Elevator Company
Ver-Trans Elevator Company
Thyssen- Krupp Elevator Company

Request for approval for equipment manufactured by any others shall be submitted in writing and shall include complete specifications of the proposed equipment specifying, building changes necessary to accommodate the proposed equipment and any other information required by the Project Administrator to insure compliance with Bidding Documents.

Approval will not be given to any Elevator Contractor who has established on prior projects, either Government, municipal, or commercial, a record for unsatisfactory elevator installations, or has repeatedly failed to complete contracts awarded to him within the contract time, or has not the requisite record of satisfactorily performing elevator installations of similar type and magnitude.

A mandatory site visit is required for the Bidders on this project to familiarize themselves with all aspects of this project. Bidders may call for an appointment to visit the job site with the Project Administrator. Name and phone number of the Project Administrator is on page 6 of these documents.

1.17 PREPARATION WORK: Preparatory work included as part of the project:

A. A properly framed and enclosed legal hoistway, including hoisting beams, ventilation, and pit with ladder, sump pump, lights and waterproofing as required by the governing code or authority.

B. All electric power for light, tools, hoists, welding. Etc., during erection, shall be available to the Elevator Contractor..

C. A suitable machine room with legal access and ventilation with a reinforced concrete structural slab which complies with the Elevator Contractor's detail template and is designed to support the impact forces indicated on the elevator layout(s). Elevator Contractor shall provide a template indicating block outs and penetrations in the machine room slab or wall, all of which must be precisely followed. All concrete work shall be coordinated to accommodate elevator equipment supplied.

D. The machine room shall be maintained at a temperature between 50 degrees F and 100 degrees F to be measured 6 inches above the floor and 1' out from any part of the car controllers, drives and motors. Areas near the heat exhaust of the controllers, drives and motors may be exempted from this temperature requirement. Relative humidity shall not exceed 95% non-condensing. Ventilation shall comply with Elevator Contractor's release requirements and Building Code. Machine room door shall be self-locking and self closing.

E. Floor Finish at Landings: A level finished floor, continuous for full width of the hoistway landing openings.

F. Location of support in overhead for the elevator beams as required by Elevator Contractor. Beam pockets required for elevator beams and patching pockets after beams are set in place.

G. Adequate supports and foundations to carry the load of all equipment, including supports for guide rails, rail bracket supports, bracket spacing as required by governing code, where required. Provide separator beams where required.

H. Vertical surfaces of entrance sill supports must be plumb, one above the other, and square with the hoistway. Finished floor and grout, if required between doorframes to sill line. For non-flush hoistways, adequate support for sill angles across full width of hoistway at each landing.

I. Hoistway walls shall be designed and constructed in accordance with required fire rating including where penetrated by elevator fixture boxes and to include adequate fastening to hoistway entrance assemblies. Front entrance walls shall not be constructed until after all elevators material is located in the hoistway. Front entrance walls shall not be constructed until after doorframes and sills are in place.

J. Elevator Contractor shall provide and install inserts, anchors, bearing plates, beams, brackets, supports, and bracing including templates and diagrams.

K. Elevator Contractor shall furnish any required concrete inserts and similar anchorage devices for the installation of guide rails, machinery, and other components of elevator work where installation of devices is indicated or required.

L. All cutting, including cutouts to accommodate hall signal fixtures, patching, and painting of walls, floors, or partitions together with finish painting of entrance doors and frames. Setting of anchors and sleeves.

M. Proper trenching and back filling for any underground piping or conduit. Grouting around the jack to be level with the pit floor as required.

N. General Contractor shall provide recesses, cutouts, slots, holes, patching, and refinishing, to accommodate the installation of all new equipment. All new holes in concrete shall be core drilled. Where concrete beams, floor slabs, or other building construction protrude more than 2" into hoist-way, all top surfaces of projections shall be beveled with 20 gauge steel to an angle of at least 75 degrees with the horizontal.

O. Clearance around elevator, mechanical and electrical equipment shall comply with applicable provisions of NEC (NFPA 70). Arrange equipment in machine room so that major equipment components can be removed for repair or replacement without dismantling or removing other equipment in same machine room. Locate controller near and visible to its respective power unit and disconnect switch.

P. Elevator Contractor shall provide sheet-metal and angle-iron guards at all openings in the machine room floor to protect personnel from accidental contact with the ropes. Openings and recesses in the machine room floor shall be covered with sheet metal or filled with concrete. The machine room floor shall be substantially flush, smooth and void of any tripping hazard.

Q. General Contractor shall caulk around and seal all piping and duct penetrations through building members, hoist-way enclosure, machine room enclosure, and ceilings with non hardening sealant.

R. ELEVATOR CONTRACTOR shall be responsible for the proper location and size of bucks, grounds and chases.

S. The smoke detector system. Elevator Contractor shall wire into the new controller the smoke detectors.

T. The Elevator Contractor shall provide a CVT (Computer Video Terminals) in the Elevator Machine Room.

U. The General Contractor shall provide guarding and protecting the hoist-way during construction. The

protection of the hoist-way shall include removable solid panels surrounding each hoist-way opening at each floor and in the machine room with a 60" high minimum guard. Hoist-way guards shall be erected, maintained and removed by the General Contractor.

V. The General Contractor shall cut all openings as required for the work under this specification including all patching and patch work for fixtures, signals, remote video terminals, hall signal fixtures, door frames, sills, machine beams, cable openings, ducts, conduit, and etc. Elevator Contractor shall furnish all necessary sleeves, thimbles, hangers, inserts, etc., at such time and in such a manner so as not to delay or interfere with the work of other trades. No beams or joists shall be cut.

W. **NEII STANDARDS FOR HANDICAPPED:** Except as otherwise indicated, comply with NEII "Suggested Minimum Passenger Elevator Requirements for the Handicapped" including clearances, handrails, locations for signal equipment and similar provisions.

X. Elevator Contractor shall provide fire extinguishers in the machine room during construction.

Y. A fused lockable disconnect switch or circuit breaker for each elevator, per the National Electrical Code . Size to suit elevator power characteristics.

Z. A separate, 120 volt A.C., 20amp, single phase power supply with SPST fused lockable disconnect switch and feeder wiring to each controller of the group for elevator signal system. Disconnect switch(es) are to be located in the elevator machine room and installed in accordance with NFPA 70.

AA. A 120 volt, 20amp, single phase power supply with SPST lockable fused disconnect switch for circuit breaker for each elevator, in compliance with the National Electrical Code, with feeder wiring to each controller for the in car lighting and ventilation fan. The disconnect switch(es) are to be located in the elevator machine room.

BB. Suitable lights and convenience outlets in the machine room with light switches located with in 18" of lock jamb side of the machine room door.

CC. A smoke and heat detector system, located as required, with wiring from smoke and heat detectors to machine room for each group of elevators. Each group of elevators requires a set of contacts NORMALLY OPEN- COMMON- NORMALLY CLOSED from the smoke detector at the designated firemen service landing, and one set of contacts NORMALLY OPEN-COMMON-NORMALLY CLOSED representing all other smoke detectors in the system.

If sprinklers are installed in the hoistway, machine room or machinery spaces, a means must be provided to disconnect automatically the main line power supply of the affected elevator prior to the application of water.

Smoke detectors shall not be used to activate sprinklers in hoistways, machine rooms or machinery spaces or to disconnect the main line power supply.

DD. For remote elevator control/indicator panels (when provided), wiring and conduit to panels located away from the elevator hoistway.

EE. The Elevator Contractor shall connect the Emergency communication connection to the elevator controller.

FF. Provide emergency power with automatic time delay transfer switch and auxiliary contacts with wiring to the elevator controller, as required.

GG. During Installation, the elevator platform(s) must not be used to hoist material or men other than required for elevator installation.

HH. Temporary use of Elevators: Should any elevator be required for use before final completion, others shall provide without expense to Elevator Contractor, if required, temporary car enclosures, requisite guards or other protection for elevator hoistway openings, main line switch with wiring, necessary power, signaling devices, lights on the car and elevator operators together with any other special labor or equipment needed to permit this temporary usage. The Elevator Contractor's temporary acceptance form shall be executed before any elevator is placed in temporary service, and the cost of power and operation, maintenance of the equipment, and rehabilitation of equipment shall be paid for by others.

II. Provide a fire extinguisher in the machine room.

1.18 EXECUTION

1.18.1 PREPARATION: Take field dimensions and examine conditions of substrates, supports, and other conditions under which this work is to be performed. Do not proceed with work until unsatisfactory conditions are

corrected.

1.18.2 SPACE CONDITIONS: Attention is called to overhead clearance, pit clearances, overall space in machine room and construction conditions. Addition or revision of space requirements or construction changes that may be required for complete installation of the elevators must be arranged for the obtained by CONTRACTOR, subject to approval of ENGINEER. Cost of such changes shall be included in bid and form a part of contract.

Clearance around elevator, mechanical and electrical equipment shall comply with applicable provisions of NEC (NFPA 70). Arrange equipment in machine room so that major equipment components can be removed for repair or replacement without dismantling or removing other equipment in same machine room. Locate controller near and visible to its respective pump unit.

1.18.3 INSERTS: Furnish required concrete inserts and similar anchorage devices for the installation of guide rails, machinery, other components of elevator work where installation of devices is indicated.

1.19 INSTALLATION: Install all elevator components except as specifically provided for elsewhere by others.

1.19.1 CUTTING AND PATCHING: Cut all openings as required for the work under this specification. Patching shall be done by the craft whose involved. Furnish and install all necessary sleeves, thimbles, hangers, inserts, etc., at such time and in such a manner so as not to delay or interfere with the work of other trades. No beams or joists shall be cut.

1.19.2 COORDINATION: Coordinate all mechanical work with other phases of the work, including ceilings systems and tiles, electrical work, other mechanical work, etc.

Coordinate all shutdowns of any and all systems with ENGINEER. Submit request for shutdowns in writing at least five (5) days prior to planned shutdown, and approval or rejection of planned shutdown shall be given by ENGINEER depending on building needs. Shutdowns may have to be scheduled on Saturdays, Sundays and Holidays and no extra compensation will be allowed for this overtime work.

1.19.3 SLEEVES AND THIMBLES: Caulk around and seal all piping and duct penetrations through building members, hoistway enclosure, machine room enclosure, and ceilings with non hardening sealant. Where slabs and walls are fire -rated, sealants shall have the same fire-rating.

1.19.4 BUCKS, GROUNDS AND CHASES: Elevator Contractor shall be responsible for the proper location and size of bucks, grounds and chases.

1.19.5 CLEANING AND ESCUTCHEONS: After all work has been finally tested, clean all piping, equipment and exposed work. All piping shall be free from all obstructions. All plated and other finished products shall be thoroughly cleaned and polished.

1.19.6 PAINTING: Elevator Contractor shall paint piping, equipment, hangers, and accessories in all mechanical equipment rooms, with two (2) coats of approved paint after thoroughly cleaning. Equipment having factory finish, if damaged or scratched, shall be touched up and given one additional coat of machinery enamel.

A. Elevator pump unit, motors, power unit, and the like shall be factory painted with manufacturer's standard finish and color.
B. Surfaces (except contact surfaces or working parts) of elevator items such as controller, car frames, underside of platform, guide rails, rail brackets, all uncoated ferrous metal items, and hydraulic piping shall be given zinc chromate or approved equal prime coat.
C. Upon completion of installation and prior to final inspection, all equipment shall be thoroughly cleaned of grease, oil, cement, plaster, and other debris. All equipment, except that otherwise specified as to architectural finish, shall then be given two coats of paint of color approved by ENGINEER.
D. Paint floor designation numbers not less than four-inch high, on hoistway doors, fascias and/or walls as required by CODE. The color of the paint used shall contrast with the color of the surfaces to which it is applied.
E. Fascia plates, toe guards, dust covers, hanger covers, and other metal shall be thoroughly cleaned and

given one field coat of paint of color approved by ENGINEER.

F. Interior and exterior steel surfaces of Elevator Cabs parkerized or given equivalent rust resistant treatment before finish is applied.

G. Interior surfaces of Elevator Cabs shall be factory finished with one coat of baked on enamel or poxylin lacquer.

H. Exterior faces of Elevator Car doors shall be given one finish with one coat of oil base paint approved by ENGINEER.

I. Elevator pump unit, controller disconnect switch and crosshead of car shall be identified by 4 inch high numerals and letters located as directed by ENGINEER. Color of number shall contrast with color surfaces to which they are applied.

1.19.7 WELDINGS at the project sits shall be made by welders and welding operators who have previously qualified by test, as prescribed in American Welding Society Publication AWS D1.1 to perform the type of work required.

1.19.8 NAMEPLATES: Each major item of equipment shall have the manufacturer's name, address, type, or style, model or serial number, catalog number, and electrical and mechanical characteristics on a plate secured to the item of equipment.

1.20 TOOL AND PARTS CABINET: Elevator Contractor shall provide as follows:

A. A metal cabinet, having two shelves and a hinged door shall be provided in the machine room. Cabinet shall measure not less than 48 inches high, 18 inches wide, and 15 inches deep. Cabinet door shall contain cylinder lock.

B. Locate the tool cabinet in the elevator machine room as directed by the ENGINEER.

Part 2. PRODUCTS:

2.1 PRODUCTS, DELIVERY, STORAGE AND HANDLING: Take necessary precautions to protect all material, equipment, apparatus and work from damage. Failure to do so to the satisfaction of the ENGINEER will be sufficient cause for rejection of material, equipment or work in question. CONTRACTOR is responsible for the safety and good condition of the materials installed until final acceptance by ENGINEER.

2.2 MATERIALS:

A. Where stainless steel is specified, it shall be corrosive resisting steel, ANSI Type 302/304 with a # 4 (satin finish) or # 8 (polished finish) on exposed surfaces to comply with ASTM a 167. Stainless steel shall have the grain of belting in the direction of the longest dimension and surfaces shall be smooth and without waves. During erection, all stainless steel surfaces shall be protected by suitable material.

B. Where cold rolled steel is specified, it shall be low carbon steel rolled to stretcher leveled standard flatness, complying with ASTM A366 cold rolled steel, commercial quality, Class 1, matte finish, stretcher leveled.

C. Where aluminum is specified , it shall be extruded per ASTM B2221; sheet and plate per ASTM B209.

D. Where plastic laminate is specified, it shall be the high pressure type complying with NEMA LD3, Type GP-50 (0.50 inch nominal thickness); color, texture, and pattern as selected by ENGINEER from standard products available in the industry. Where fire retardant particle board panels are specified, it shall be minimum 3/8 inch thick back up for plastic laminate panels, edged and faced, provided with suitable anti-warp backing; meet ASTM E84 Class "I" with flame spread rating of 25 or less.

E. Where resilient rubber tile is specified; FS SS-T-312, Type III, homogeneous (solid) vinyl tile, Class 1, 12 inch by 1/8 inch thick; color and pattern selected by ENGINEER from manufacturer's standard ranges.

F. Bronze: 60% copper and 40% zinc in alloy 280. Finishes in # 4 (satin finish) or # 8 (polished finish).

G. Baked Enamel: Electro-statically applied, oven baked to match selected color.

H. Natural Wood Veneer: Plain sliced, book matched, premium grade.

I. Carpet: ASTM E 648, Class I rating, type I, level loop, solution dyed.

J. Glass: Clear laminated and meet the requirements of ANSI Z 97.1.

K. Clean all exposed metal of oil, grease, scale and other foreign matter and factory paint on shop coat of manufacturer's standard rust resistant primer. After erection, provide one finish coat of industrial enamel

paint. Galvanized metal need not be painted.

L. Grout: Non-shrink, Non-metallic Grout: Factory packaged, non-staining, non-corrosive, non-gaseous grout complying with ASTM C 1107. Provide grout specifically recommended by manufacturer for grouting door sills and similar applications.

2.3 **HYDRAULICS AND ELEVATOR EQUIPMENT:** Provide the following manufacturer's standard elevator systems.

A. Conventional Hydraulic System: Complete elevator system consisting of single/multiple stage-acting under-the-car hydraulic plunger- cylinder unit requiring minimal depth cylinder drilling/excavation. Elevator installation shall be complete with electric pump-tank-control system equipment in machine room as indicated.

B. MOTOR STARTER: Elevator Contractor shall provide a solid state starter to limit current in-rush during starting and to provide gradual acceleration of the motor. Motor starting shall not be initiated by mechanical contacts. The starter shall include a current limit adjustment range of 200% to 450% of the overload adjustment range. An integral fault detection and diagnostic system shall be provided. It the internal fault detection system detects a failure, power shall be removed from the motor.

2.4 **PIPING:**

A. Provide size, type, and weight piping recommended by manufacturer, and provide isolation couplings to prevent sound /vibration transmissions from power unit. Connections shall be furnished and installed between the storage tank, pump, muffler, operating valves, and cylinder complete with necessary valves, pipe support and fittings. All connections between the discharge side of the pump, check valve, muffler, cylinder, lowering valves shall be of schedule 80 steel with screw, flanged, welded, or approved flexible or mechanical couplings.

B. Provide supports for all horizontal piping. Place hangers or supports within 12-inches on each side of every directional change of pipe line and space supports not over 10 feet apart. Secure vertical runs properly with iron clamps at sufficiently close intervals to carry weight of pipe and contents. Provide supports under pipe to floor and isolate hanger.

C. Install pipe sleeves where pipes pass through walls, floors, etc. Sleeves shall be set during construction. After installation of piping, the sleeves shall be equipped with snug fitting inner liner of either glass or mineral wool insulation.

2.5 **SPECIAL TOOLS:** One set of special tools, calibration devices, diagnostic devices and tools, and instruments required for operation, calibration, adjustment and maintenance of the equipment shall be provided and become the property of Southeastern's Physical Plant. The diagnostic devices shall be equipment to be maintained by the ELEVATOR CONTRACTOR in accordance with the Elevator Maintenance Contract upon expiration of the warranty period.

2.6 **DESCRIPTION OF EQUIPMENT: ONE NEW PASSENGER ELEVATOR**

A. Control:	Microprocessor based, integrated solid state circuitry and as specified.
B. Capacity:	3,500 lbs
C. Speed:	125 fpm
D. Operation System:	Selective-collective Automatic Operation. Single car
E. Travel:	+/- 15 feet verify prior to fabrication
F. Power Supply:	Power Supply 480V, 60 Hz, 3 Phase. (Contractors must verify what is available in the building.)
G. Machine Location:	adjacent to shaft at first floor
H. Stops:	(02) One landing. Normal Operation: In line. Emergency Operation. Return to the first floor.
I. Openings:	(01) one car opening (verify)
J. Door Opening Operation:	Single speed side opening. 42" by 82" Automatic. (See plans)
K. Power unit, oil storage tank, oil control unit:	Provide new as specified.
L. Car Enclosure:	Provide new. Inside dimensions: Width: 80 inches

Depth: 65 inches: Inside height: 96 inches. Front walls: # 4 Stainless Steel with integral car door frames. Side and Rear Walls: 16 ga. Enameled Steel panels ith raised plastic laminate panels. Car fixtures: Satin stainless steel.

Door Faces (Interior): Satin stainless steel. Flooring prepared to receive resilient tile (specified in Division 9 Section "Resilient Tile Flooring"). Provide protective blanket hooks in the car and one set of full height blankets.

M. Car platform, car frame and car sling: Provide new. As specified. Platform dimensions to measure.

N. Signal Equipment: As specified.
Illuminated vandal resistant car and hall push buttons
Discrete car position indicator
Car riding lantern and audible gong
Car fixtures: # 4 stainless steel and as specified.
Hall fixtures: # 4 stainless steel and as specified.

O. Special Features: Independent Service
Limited door reversal
Handicapped features as per ADA Requirements
Firefighters Service Phase I & II per ASME A17.1 Rule 211.3
Vandal resistant speaker telephone meeting ADA requirements
Braille markings and signage

P. Buffers: Provide new spring buffers as specified.

Q. Finishes: As specified and as approved by ENGINEER.

R. Handrails 3/8" by 2" # 4 stainless steel on rear wall.

S. Ceiling Enameled steel with 2 - 2' recessed fluorescent, vandal resistant fixtures

T. Hoistway entrances: Hoistway Entrances: As follows:
Width: 42 inches.
Height: 84 inches.
Type: Single-speed (single) or double speed (double door) side sliding.
Frames: Satin stainless steel.
Doors: Satin stainless steel
Sills: Aluminum

U. Hall fixtures: Satin Stainless steel vandal resistant

CONTRACTOR shall deliver to job site in manufacturer's original, non-broken, unopened factory packaging, labeled with manufacturer's density and thickness. Store materials in original protective packaging. Prevent soiling, physical damage or wetting.

2.7 DOOR OPERATOR, DOOR PROTECTIVE DEVICES, AND DOOR OPERATIONS:

2.7.1 DOOR OPERATOR: The door operator shall be microprocessor based and reside in the door operator controlling all functions of the door. The microprocessor door operator shall be linked to the main microprocessor through a serial communication link. A new direct current, motor driven heavy-duty, high speed operator shall be furnished and installed that will automatically open the car and hoistway doors simultaneously when the car is leveling and automatically close the doors simultaneously at the expiration of the door open timing. The direct-current motor shall be of the high internal resistance type, capable of withstanding high currents resulting from stall without damage to the motor. The door operator shall be capable of opening a car door and hoistway door simultaneously at a maximum speed of not less than 2.0 feet per second. Door movements shall be electrically cushioned at both limits of travel. Reversal of direction of the doors from the closing to opening operation, whether initiated by the electronic door reopening device or the door "OPEN" button, shall be accomplished within no more than 1-1/2 inches maximum of door movement. Doors shall operate smoothly without vibration, slam, or shock. The motor shall conform to NEMA MG 1, and be UL listed. Direct drive geared operators, AC controlled units with oil

checks, or other deviations from the above are not acceptable.

Door restrictive devices shall be provided which meet all requirements of A17.1 Rule 111.12. If and only if the car is within the landing zone, when the electric power fails or is interrupted, the door operator shall permit the doors to be opened readily by hand from within the car. Direct drive geared operators, A.C. controlled units with oil checks, or other deviations for the above are not acceptable.

The door operation shall be integrated with car leveling system to prevent elevator movement before the doors close. The elevator shall maintain a self leveling feature that will automatically bring the car to the floor landing. This self leveling shall, within its zone, be entirely automatic and independent of the operating devices and shall correct for over travel or under travel. The car shall also be maintained approximately level (within + or - 1/8") with the landing irrespective of the load.

2.7.2 DOOR CLUTCHES, LINKAGE ARMS, DRIVE BLOCKS AND CAMS: Furnish and install heavy duty clutches, linkage arms, drive blocks, pickup rollers and cams to provide positive, smooth, quiet door operation. Design clutches so car doors can be closed for maintenance purposes while hoistway doors remain open.

2.7.3 DOOR SAFETY DEVICES: Install new door protection system using 40 microprocessor controlled infra-red light beams, calibrated or adjusted to meet the following requirements. The beams shall project across the car opening detecting the presence of a passenger or object. If door movement is obstructed the doors shall immediately reopen. The door shall automatically open when the car arrives at the landing and shall automatically close after an adjustable time interval or when the car is dispatched to another landing. Closing may be initiated instantaneously be registration of a car call, operation of load weighing device or signal from the service demand integrator.

If the electronic detector is activated when the doors are closing and the doors are more than one-third closed, they shall reverse direction and open only partially. The doors shall begin to reclose when the electronic detector is deactivated. The doors shall reopen fully if the electronic detector is activated longer than a fixed interval.

If a passenger or object is detected during normal closing operation, the doors will immediately stop and reopen. Closing will be initiated one-half second after the passenger or object has been removed from the opening.

The doors will remain open as long as the electronic detector senses the presence of a passenger or object in the door opening. If door movement is obstructed longer than a field programmable time value, a buzzer will sound and the doors will close at a reduced speed (go into nudging speed). If the infra-red door protection system detects a person or object when closing, the doors will stop and resume closing when the obstruction has been removed.

The current door "HOLD OPEN" time shall be changed to a shorter field programmable time when the door protection system is activated. The microprocessor control system shall provide separate timers for car call door "DOOR OPEN" time and hall call "DOOR OPEN" time. The door hold open times shall be field programmable.

2.7.4 DOOR TIMING DEVICE: Provide a timing arrangement for door operation which is adjustable to conform with various types of traffic in and out of the elevator. Include the following requirements.

A. NORMAL OPERATION: When an elevator stops in response to a landing call, the door open time shall be lengthened sufficiently to give the first waiting passenger time to reach the opening before the door closes. Door open time shall then be reduced to accommodate all subsequent passengers. When an elevator stops in response to a car call, the doors shall start to close immediately after the first passenger or a stream of passengers has left the car.

B. NUDGING OPERATION: Provide nudging operation so that if, for any reason, the doors are held open longer than a predetermined period, either by someone standing in the opening, holding the electronic door detector, or the door open button, the doors shall close at slow speed with the electronic

detection system cut out and a buzzer shall operate. Should the door detector encounter a passenger during nudging, the doors shall stall and remain in the stalled position until released. When the obstruction is removed, the doors shall resume normal speed.

C. DOOR OPEN SENTINEL: If the doors are opening but do not fully open after a field adjustable time, the doors will close then recycle up to six times to try and correct the fault.
D. DOOR CLOSE SENTINEL: If the doors are closing but do not fully close after a field adjustable time, the doors will open then recycle up to six times to try and correct the fault.
E. DOOR CLOSE ASSIT: When the doors have failed to fully close and are in the recycle mode, the door drive motor shall have increased torque applied in an effort to possibly overcome any mechanical resistance or differential air pressure and allow the door to close.

2.8 PASSENGER CAR DOORS, HOISTWAY DOORS, ENTRANCES, CAR-HOISTWAY DOOR ACCESSORIES:

2.8.1 CAR DOORS: Provide car doors for entrance to the car. Provide width and height of the opening not less than that provided by the hoistway doors. The car doors shall be the same type as the hoistway doors. The car door panels shall be provided with sound deadening material and have 1-1/2 hour UL rating.

Equip doors with infra-red electronic door protective device. The car side of door finish shall be 0.30" thick plastic laminate. The doors shall be hung on sheave type hangers with polyurethane tires that roll on a polished steel track. Bottom of each door panel shall be guided by two non metallic gibs on each panel. Gibs shall be replaceable without removing the door from the hanger rail. Door finish shall be # 4 Satin finish stainless steel.

2.8.2 HOISTWAY ENTRANCE DOORS: ASME A17.1 Rules 110.11e and 110.11f; and label for 1-1/2 hour fire rating. Provide hollow metal non vision construction, of plain panel design, with flush surfaces on the car and landing sides. The door panels shall be not less than 1-1/4 inches thick, of minimum 16 gauge flush steel construction and shall be provided with 16-gauge sight guard to match the finish door. Each door shall be reinforced inside with continuous vertical members and shall be sound deadened. Doors shall be reinforced to accept the required operating mechanism and hardware. No seams, binding strips, or screws shall be visible from the landing. Finish shall be _____# 4 Satin finish stainless steel as approved and selected by ENGINEER.

A. The doors shall be hung on sheave type hangers with polyurethane tires that roll on a polished steel track.
B. Guide the bottom of each panel with at least two nonmetallic gibs, sliding in a machined groove in the landing sill. Gibs shall be replaceable without removing doors from hanger rails. Provide door panels with stainless steel mop strips.
C. Install two door jamb Braille plates per entrance.
D. Door restrictive devices shall be provided and installed which meet or exceed the requirements of ASME A17. 1 Rule 111.12.

2.8.3 CAR DOOR AND HOISTWAY DOOR HANGER SUPPORTS:ASME A17.1 RULE 110.11B. Hangers for the car doors and hoistway doors shall

A. Each door panel shall have not less than two hangers per door panel.
B. The hangers shall be of two point suspension type, consisting of nylon or other non-metallic tired sheaved fitted with dust proof, grease asked ball or roller bearing mounted on a malleable iron or steel bracket.
C. The door rollers shall be made of, or tired with, a sound reducing material, and shall rotate on grease packed ball bearings. The diameter of the rollers shall not be less than 3-1/4" for car doors and not less than 2-1/4" for hoistway doors.
D. The hanger sheaves shall operate at a relatively low rotational speed and shall roll on a new high carbon, cold rolled or drawn steel track shaped so as to permit free movement of sheaves without regard to vertical adjustment of sheave, bracket, or housing.
E. The hanger track shall be of high carbon cold-drawn steel not less than 2" high and 2" thick, and shall be rounded at the top to receive the door rollers, and at the bottom to receive the up-thrust rollers, or as approved by ENGINEER.
F. Beneath the track and each hanger sheave, a hardened steel up-thrust roller shall be provided capable of withstanding a vertical thrust equal to the carrying capacity of adjacent upper sheave. The up-thrust shall have fine vertical adjustments, and the face of the roller shall be shaped so as to permit free

movement of the hanger sheave. The up-thrust roller shall have ball or roller bearings assembled on an eccentric stud.

G. The hanger sheaves shall be provided with steel fire stops to prevent disengagement from tracks.

H. Each hanger sheave shall be fitted with a felt, or similar material, wiper to provide a light lubricating film between the sheave and hanger track.

I. Hangers that are constructed integral with the door panels shall not be used.

2.8.4 HOISTWAY DOOR CLOSERS:ASME A17.1 RULE 110.3. Provide spring, counterweight or spirator closers to ensure positive closing of the hoistway doors.

2.8.5 HOISTWAY DOOR INTERLOCKS AND CAR DOOR CONTACTS:Equip each hoistway entrance with approved type interlocks. An electrical interlocking device separate from mechanical interlock will not be approved. The interlock shall prevent operation of the car until the hoistway are locked in the closed position.

The interlock shall also prevent opening the doors at any landing from the landing side except by special key, unless the car is at rest at that landing, or is in the leveling zone and stopping at the landing. Accidental ground in the electric circuit shall not prevent the operation of the interlocks, Locate the interlocks to provide a ready, unobstructed access for periodic maintenance, inspection, and repairs. Equip car doors with an electrical contact which will prevent operation of the car unless the doors are closed. Interlocks shall bear Underwriters Laboratories "B" label of approval.

2.8.6 HOISTWAY DOOR UNLOCKING DEVICE: Hoistway door unlocking devices a specified by ASME A17.1 Code shall be provided to permit authorized persons to gain access to the hoistway when the elevator car is away from the landing.

2.8.7 HOISTWAY ENTRANCE FRAMES: ASME A17.1 RULE 110.11c. Fabricate of steel sheet with minimum 14 gauge, with baked enamel finish as selected and approved by ENGINEER. The component members constituting the unit frame shall include header, struts, and frame of door opening. The head and jamb section shall be bolted assembly with bolts, washer and locking nut or lock washer. The frame shall be securely fastened to the structure. For concrete walls, provide full length knockdown back flanges. For other type walls, provide three suitable anchors on each side of frame. Paint inside of The frame with 3/16 inch thick sound isolating material. Reinforce the frame more than 8 inches wide. Grout the frame after installation. Frame finish shall be # 4 satan stainless steel.

A. ALIGNMENT: Coordinate installation of hoistway entrances with installation of elevator guide rails for accurate alignment of entrances with cars. Where possible, delay installation of sills and frames until car is operable in the shaft. Reduce clearances to minimum, safe, workable dimension at each landing.

B. HOISTWAY ENTRANCE DESIGNATIONS: Identification Plates: Each hoistway entrance shall have a raised and Braille floor designation provided on both jambs. Mount such that center lines of characters, shall be 60 inches above the finished floor. Characters shall be 2 inches high and shall comply fully with Section 4.30.4 of the Americans With Disabilities Act Legislation.

2.8.8 HOISTWAY ENTRANCES SILLS: ASME A17.1, Rule 110.11a, one piece cast or extruded aluminum, with grooved, non slip surface, mill finish, extending from strut or strut. Securely fasten the entrance sill to floor or sill supports. No fastening screws shall be visible. Machined grooves for door guides shall be not less than 2 inch wide. Grout the sill solidly its full length after installation with approved grout materials.

Set sills flush and level with finished floor surface at landings. Coordinate with other trades to facilitate and ensure proper grouting of sills. Fill space under sill solidly with non-shrink, nonmetallic grout.

2.8.9 STRUT ANGLES: Strut angles, when required, shall be of not less than 3 by 3 by 3/16 - inch structural steel extending from the sill to the beam above and securely anchored to building construction. Fastenings and bracing shall be with structural members having a cross section of not less than the strut angles.

2.8.10 COVERS: Hanger covers, dust covers, toe guards, and fascia plates shall be a minimum of 16 gauge

reinforced galvanized steel, and finished with one prime coat and one field coat of approved paint.

A. HOISTWAY ENTRANCE SUPPORT HANGER COVERS: Provide full length entrance door hanger covers constructed to permit ready access for servicing the door hangers. The cover shall be removable in three or more sections. Fasten the dust cover to the door hanger housing and building construction.
B. HOISTWAY ENTRANCE FASCIA: ASME A17.1, Rules 110.10a and 110.11a. Fascias shall be provided between each door hanger housing and the sill above.
C. TOE GUARDS: Toe guards shall be securely fastened to the sills.

2.9 OPERATING AND CONTROL SYSTEMS: SINGLE CAR SELECTIVE COLLECTIVE AUTOMATIC OPERATION:AUXILIARY OPERATIONS: SECURITY FEATURES

2.9.1 AUTOMATIC PUSH-BUTTON OPERATION: Operate elevator without an attendant from buttons located at each floor and in the car. When car is idle, automatically start car and dispatch it to floor corresponding to registered car or hall call. The elevator control shall be a distributed control system, microprocessor based and software oriented. The main microprocessor and car controller can be located behind the elevator swing return panel or in the elevator machine room. The microprocessor selector, situated on the car top and the microprocessor door operator, residing in the door operator shall be linked together with main processor by a serial communications link. Control of the elevator shall be automatic in operation by means of push buttons in the car numbered to correspond to floors served, for registering car stops by "up-down" push buttons at each intermediate landing and "call" push buttons at terminal landings.

The momentary pressing of one or more buttons shall dispatch the car to the designated landings in the order in which the landings are reached by the car, irrespective of the sequence in which the buttons are pressed.

Each landing call shall be canceled when answered. When the car is traveling in the up direction, it shall not stop at floors for which car buttons or "up" hall buttons have been pressed; it shall not stop at floors when "down" buttons only have been pressed, unless the stop for that floor has been registered by a car button, or unless the down call is at the highest floor for which any buttons have been pressed. Likewise, the pressing of an "up" down when the car is traveling in the down direction shall not intercept the travel unless the stop for that floor has been registered by a car button, or unless the up call is the lowest for which any button has been presses.

When the car has responded to its highest or lowest stop, and stops are registered for the opposite direction, its direction of travel shall reverse automatically and it shall then answer the calls registered for that direction.

Should both up and own calls be registered at an intermediate floor, only the call corresponding to the direction in which the car is traveling shall be canceled upon the stopping of the car at the landing.

An adjustable time delay shall be provided so that after the car has stopped in response to a hall button, the entering passenger may register his car button before the car will reverse to answer calls in the opposite direction.

A. The elevator shall be served by indicated push-button riser(s) at each landing.
B. Car buttons shall be marked for corresponding landings.
C. Minimum diameter of each button shall be ¾ - inch in the smallest dimension.
D. All buttons in the elevator and hall call stations are to be Vandal Resistant type.

2.9.2 MICROPROCESSOR CONTROL SYSTEM:

A. Provide solid state components and printed circuit boards to control the power unit and signal functions in accordance with these specification. Complete details of the components and printed circuit boards, together with a complete operational description, shall be submitted for approval.
B. All controller assemblies shall provide efficient, smooth, stepless acceleration and deceleration of the elevator, automatically and irrespective of load in the car. All control equipment shall be enclosed in a metal cabinet with lockable, hinged, door(s) and shall be provided with a means of ventilation. All non conducting metal parts in the machine room shall be grounded in accordance with the NEC.

Cabinet shall be securely attached to the building structure.

C. Modules for the control of the elevator system, including dispatching, signals, door operation and special operation, shall be installed in a NEMA, Type 1, General Purpose Enclosure. Circuit boards shall be moisture-resistant material and of adequate thickness to support the components mounted thereon.

D. Each device, module and fuse (with ampere rating) shall be identified by name, letter or standard symbol in approved indelible and legible manner on the device or panel. Coordinate identification markings with identical markings on wiring diagrams.

E. Modules shall be keyed or notched so as to prevent insertion of the modules in the inverted position.

F. Light emitting diodes (LEDS) shall be for visual monitoring of individual modules.

G. Components shall have interlocking circuits to assure fail-safe operation and to prevent unwarranted elevator movement should any component fail to function properly.

H. Modules shall be of the type that plug into pre-wired mounting racks. Field wiring or alteration shall not be necessary in order to replace defective modules.

I. All logic symbols and circuitry designations shall be in accordance with ASME A17.1 Standards.

J. Solid state components shall be designed to operate within a temperature range of 0 degrees and 110 degrees F. No temperature controller or air-conditioned rooms shall be required for proper operation of solid state components.

K. Wiring connections for operating circuits and for external control circuits shall be brought to terminal blocks mounted in an accessible location within the controller cabinet. Terminal blocks using pierce-through serrated washers shall not be acceptable.

L. Microprocessor Related Hardware:

1. Include built-in noise suppression devices, which provide a high level of noise immunity on all solid state hardware and devices.
2. Provide power supplies with noise suppression devices.
3. Isolate inputs from external devices (such as push buttons with opto-isolation modules).
4. Provide separate regulated power supply for each computer chassis.
5. Design control circuits so that one side of power supply is grounded to provide for testing.
6. Under no circumstances, shall the safety circuits be affected by accidental grounding of any part of the system.
7. Design system so that it will start properly when power is restored in the event of a power failure or interruption.
8. Provide non-volatile system memory so that data is retained in the event of power failure or disturbance.

M. Provide UL or CSA recognized power supplies with short circuit protection.

N. Use stable capacitor or crystals as the time basis for electronic time delay devices.

O. Permanently mark components (relay, DC boards, etc.) with symbols shown on Drawings.

P. The following information relating to the specific type or microprocessor controls installed on this project shall be provided at the time of submittal:

1. Owner's information manual, containing general data on major components ,maintenance and adjustments.
2. System logic description.
3. Complete wiring diagrams needed for field trouble shooting, adjustments, repair and/or replacement of components.
4. Diagrams shall be base diagrams.
5. Device program tapes.
6. Base software listing.
7. Password or Code to access control equipment.

2.9.3 POSITION SELECTOR: The position selector shall be part of the microprocessor system. The composition in the hoistway shall be digitized through a primary position encoder. The microprocessor control system shall store floor position and slowdown points in memory.

2.9.4 LEVELING DEVICE: An automatic two-way leveling device shall be incorporated so that the car shall approach landing stops at reduced speed from either direction of travel, and automatically level at landings with an accuracy of plus or minus ¼ inch under varying loads. Landing level shall be maintained as long as the car is within the leveling zone. Terminal stopping devices shall be provided as required by ASME A17.1

2.9.5 CAR TOP INSPECTION STATION:

A. The device shall conform to the CODE and the following:
B. The device shall be activated by a toggle switch mounted in the device. The switch shall have the "Automatic" and "Inspection" positions permanently marked the face plate with ¼ inch letters.
C. Movement of the elevator shall be accomplished by the continuous pressure on a direction button and a safety button.
D. Provide an emergency stop toggle type switch as per CODE.
E. Provide permanent identifications for the operation of all components in the device.
F. The device shall be permanently attached to the elevator crosshead on the side of the elevator which is nearest to the hoistway doors.

2.10 CAR STATION PANELS, POSITION INDICATORS, HALL LANTERNS, SIGNALS, DESIGNATIONS, HALL CALL OPERATION STATIONS:

2.10.1 CAR STATION PANELS:The car station shall contain the devices required for The specified operations. The emergency stop switch shall be keyed and red in color. The car floor register buttons shall be metal encased, stainless steel, vandal resistant type an embossed to permit illumination when a call is registered and will remain illuminated until the call is answered. Buttons shall be designated to seat on the face plate in lieu of the button mechanism, and with 1/32 - inch operating clearance. Buttons shall have maximum protrusion of 3/16 - inch beyond the face plate, with beveled edges to prevent damage from side blows. All raised floor indications and handicap symbols (Braille), shall be located immediately adjacent to the floor buttons and fully integrated in the module design. No applied symbols shall be allowed. There shall be no floor indications or symbols on the buttons. "NO SMOKING ALLOWED" shall be engraved on the upper portion of the car station with numerals 1-inch high.

Car station panel shall consist of a series of modules all designed for optimum viewing and accessibility. The lowest module shall contain the "DOOR OPEN", "DOOR CLOSE", "EMERGENCY STOP SWITCH", and "ALARM" button. The intermediate module shall contain the floor buttons. Top module shall contain required switches.

A two position key operated "INDEPENDENT SERVICE SWITCH" shall be provided in main car operating panel which shall have its positions marked "AUTOMATIC" and "INDEPENDENT SERVICE".

When switch is in the "INDEPENDENT" position, the car shall respond only to calls registered on its car dispatch buttons and shall bypass all calls on landing push buttons. Car and hoistway door shall not close until a car button or the "DOOR CLOSE" button is pressed and held until the interlock circuits are established. When the switch is returned to the "AUTOMATIC" position, normal service shall resumed. Key for this switch shall be removable only in the "AUTOMATIC" position.

2.10.2 CAR POSITION INDICATOR: Provide and electronic Dot Matrix car position indicator. As the car travels, its position in the hoistway shall be indicated by the illumination of the Alpha/Numeric characters corresponding to the landing which the elevator is stopped or passing. The position indicator shall permit reading of numerals without glare from any point in the car during normal operations. Numerals shall be a minimum of 25.4 mm high. The position indicator shall be mounted in a module matching the control panels for optimum viewing.

2.10.3 DIRECTION LANTERNS AND AUDIBLE SIGNALS: Lanterns shall be designed in accordance with ADA requirements and be provided in each car entrance column and shall be the manufacturers standard vandal resistant design.

A. Lanterns shall signal the approach of a stopping car when the car is a predetermined distance from the landing.
B. Direction lanterns in car entrance column shall include an audible signal which shall sound coincident with lantern illumination indicator.
C. Audible Signals incorporated into the elevator control annunciation system shall "Chime" once for the up direction and twice for the down direction of travel of each elevator car.

D. A second audible signal, having a sound of other than the "Gong" type, shall also be incorporated, and shall sound once each time the car passes or stops at any landing.

2.10.4 CORRIDOR POSITION INDICATOR AND DIRECTIONAL ARROWS: Provide combination hall lantern and electronic Dot Matrix position indicator with directional arrows directly over hoistway entrances at the main lobby floor. As the car travels, its position in the hoistway shall be indicated by the illumination of the Alpha/Numeric characters corresponding to the landing which the elevator is stopped or passing. Numerals shall be a minimum of 25.4 mm high. The fixture design and operation shall be similar in design to the specified for car position indicator.

2.10.5 DESIGNATIONS FOR ENTRANCES (BRAILLE) Raised stainless steel symbols as required by ADA and CODE, of color selected by ENGINEER, shall be provided at each floor to indicate the floor location. Symbols shall be attached with concealed fasteners. Symbols shall be located so as to be seen from the open elevator door.

2.10.6 FACE PLATES:

A. Face Plates for all elevator operating and signal devices shall be fabricated from not less than 1/8 inch thick flat stainless steel No. 4 satin finish, with all edges beveled at least 15 degrees. All face plates shall be installed flush with surface upon which they are mounted.
B. Corridor push-button face plates shall be at least 2 inches wide by 8 inches high. The centerline of the corridor push button fixture shall be 42 inches above the corridor floor.
C. Fasten all car and corridor operating device and signal device face plates with non-corrosive white metal spanner head or Bristol head tamper proof screws.
D. Car and corridor push-button face plates shall be designed so that pressure on push buttons shall be independent of pressure on push button contacts.
E. Engraved legends or raised numerals in face plates shall have lettering ¼ inch high filled with black paint.
F. Engrave safety message "IN CASE OF FIRE USE STAIRWAY FOR EXIT. DO NOT USE ELEVATOR" or provide similar message on plaque.
G. Each face plate shall include raised or cast tactile letters or symbols per ADA Handicapped requirements, including Grade 2 Braille.
H. Engravings shall be "Black" filled, with the exception of firefighters service identification, which shall be "Red" filled.

2.10.7 OPERATING DEVICES AT HOISTWAY LANDINGS (HALL CALL STATIONS). Locate landing call buttons so that the horizontal centerline between the buttons is 42 inches above the floor. The landing call register buttons shall be metal encased, stainless, vandal resistant type, and embossed to permit illumination when a call is registered and shall extinguish when that call is answered. Buttons shall be ¾ inch in the smallest dimension, designed to seat n the face plate in lieu of the button mechanism, and with 1/32 inch operating clearance. Buttons shall have maximum protrusion of 3/16 inch beyond the face plate, with beveled edges to prevent damage from side blows. Provide mechanically fastened Braille symbols directly to the left of call buttons. Objects mounted beneath landing call buttons shall not project in the elevator landing more than 4 inches. The buttons shall be illuminated to indicate that a call has been registered at that floor for the indicated direction.

2.11 **ELECTRICAL PROTECTIVE DEVICES:** Equip Elevators with electrical protective devices as required ASME A17.1, Rule 210.2

2.11.1 CONTROLLER AND FAILURE PROTECTION: Components required for proper performance of the elevator shall be neatly mounted, wired, and completely enclosed in a cabinet with a mechanically latched door. The control cabinet shall be designed for mounting on the power unit. The electric control apparatus shall be completely isolated form the oil reservoir. A feature shall be incorporated in the electrical control circuit which shall cause the elevator car to descend automatically to the lowest terminal landing, if the system runs low of oil during ascending of the car. The car and hoistway doors shall automatically open when the car reaches the landing to allow passengers to exit. The parked car will have

doors I closed position and all control buttons shall be made inoperative except the "DOOR OPEN" button in the car station.

2.11.2 OVER LOAD CURRENT PROTECTION: Provide equipment with overload current protective devices, one for each pole. Phase protection relay shall also be provided. Provide three phase VOLTAGE MONITER to protect against Phase Loss, Phase Reversal, Unbalanced Voltages, Short Cycling, Low Voltages and High Voltage.

2.11.3 TERMINAL STOP:

A. Mount normal terminal stopping switch on car or in hoistway as required by ASME A17.1 Rule 305. Switches shall bring car to a stop within 2 inches of the terminal floor with any load up to and including the rated load in the car.
B. Switch, when opened, shall permit operation of car in reverse direction.
C. No normal stopping device other than one mounted on car and activated by cams in the hoistway, or mounted in hoistway and activated by cams on car, shall be permitted.
D. Final stopping switches shall be independent of other stopping devices and shall, when opened, remove power from pump motor and control valves to prevent operation of car in either direction.

2.12 CAR ENCLOSURE, PLATFORM, GUIDES, GUIDE SHOES AND CAR FRAME:

2.12.1 GENERAL: Provide manufacturers standard pre-engineered enameled-steel car enclosures with non-removable wall panels, suspended security ceiling, of the selections indicated. Include ventilation, lighting, ceiling finish, wall finish, access doors, power door operators, sill (threshold), trim, accessories, access doors, and floor finish unless indicated as not work of this Section. Provide horizontal sliding doors of manufacturers standard flush panel type, with operation and number of panels as indicated. Provide manufacturers standard protective edge trim system for door and wall panels, except as otherwise indicated.

2.12.2 PASSENGER ELEVATOR CAR ENCLOSURE: ASME A17.1, Section 204.

A. Fabricate car with recesses and cutouts for signal equipment.
B. Floor finish is specified in another Section.
C. Fabricate car doorframe integrally with front wall of car. CAR FRONTS shall be combination door post and return panels manufactured from # 4 satin stainless steel and with necessary cutouts for operating devices. Operating panel shall be recessed into the front return panel with the operating panel cover surface applied. Position indicator in the door frame header shall be recessed with a surface applied cover plate. Finish shall be # 4 stainless.
D. Stainless-Steel Doors: Flush, hollow-metal construction, fabricated from satin finished stainless steel.
E. Sills: Extruded metal, with grooved surface, ¼ inch thick.
F. Stainless Steel Wall Panels: Provide heavy duty "rigidized" hardened and textured stainless steel wall panels attached to the steel walls with concealed fasteners.
G. Ceiling/Cab: Provide a fully exposed, baked enamel steel cab, walls, and ceiling, with recessed lights. Cab shall be heaviest metal thickness available from the elevator manufacturer.
H. Lights: Fluorescent light fixtures are to be recessed into the metal ceiling at the carrier beams, covered with a vandal resistant and unbreakable translucent cover attached with TORX head tamperproof screws.
I. Handrails: Provide manufacturers standard 3/8 by 2" stainless steel handrails on rear wall unless otherwise indicated either continuous or segmented units.
J. Ventilation: Drill symmetrically spaced holes in the metal ceiling in a circular pattern at the ventilation fan. Show layout for approval in shop drawings. Provide a forced supply air ventilating system incorporating a two - speed squirrel cage blower delivering the air required at a fan speed of 1600 rpm. Design mounting to minimize noise and prevent vibration. Motor shall be energy efficient type and meet requirements of NFPA 70. Provide inlet bent slots in side panels not more than 12 inches above the base. Provide a switch on the car panel to control this fan.
K. EMERGENCY EXIT Access panel from above: ASME A17.1 Rule 204.1. Provide an emergency exit in the top of the car with a secure lock on the access panel from above. Arrange the car top exit to open only from outside of the car, without the use of tools. Equip the exit with a non-self-resetting contact , which shall prevent operation of the car unless the exit panel is fully closed.

Cause alarm system to sound when the exit door opens.

L. CAR FRAME AND PLATFORM: ASME A17.1, Section 203. Provide a car frame and platform of Manufacturers standard welded steel unit. Frame shall be constructed of structural of formed steel shapes welded or bolted together, consisting of cross head and bolster with channel uprights, gusset plates and diagonal bracing.

M. Cold-Rolled Steel: Where specified shall be low carbon steel rolled to stretcher leveled standard flatness, commercial quality, Class 1, matte finish complying with ASTM A 366. Colors shall be selected by ENGINEER.

N. Stainless Steel: ASME Type 302/304 with No 4 satin finish.

O. Plastic Laminate Wall Panels: Provide high-pressure type complying with NEMA LD 3, general purpose type; color, texture, and pattern as selected by Engineer.

P. Vinyl Tile Floor Covering: FS SS-T-312, Type III, homogeneous (solid) vinyl tile, Class 1, 12 inch by 12 by 1/8 inch thick; color and pattern selected by ENGINEER from manufacturers standard ranges.

2.12.3. CAR EMERGENCY LIGHTING POWER PACKAGE AND EMERGENCY SIGNAL SYSTEM.

A. POWER PACK: The emergency signal and emergency car lighting system shall consist of an emergency power pack mounted on top of the elevator with incandescent lamp and fixtures, a remote lighting fixture inside the elevator car, located in the car operation panel. This package shall contain a rechargeable battery of the sealed gelatin electrolyte type, with solid state controls, and an integral regulating charger connected to normal power supply. This unit shall contain the following:

1. Minimum six inch diameter alarm bell connected to elevator alarm button.
2. TOP - OF - Car light fixture with ON - OFF switch and protective wire guard. Do not provide a plastic guard.
3. Testing circuit and pilot light.
4. Low wattage pilot light indication.

B. EMERGENCY LIGHT FIXTURE: Locate Emergency light fixture inside elevator car, in car station, with flush mounted lens and shall consist of the following:

1. A minimum of two lamps capable of providing a minimum level of illumination of 0.2 foot candle at a point four (4) feet above the floor, 1-foot in front of the car station.
2. Fixture frame of stainless steel and lens of frosted acrylic, ¼ inch.

C. REMOTE LIGHT FIXTURE: Upon interruption of normal power, remote light fixture shall automatically illuminate within10 seconds and permit operation of the alarm bell, subject to the activation of the alarm button.

D. The power pack shall be capable of providing a minimum of 1 hour emergency bell operation and 4 hours of continuous illumination.

2.12.4 CAR FRAME AND PLATFORM: ASME A17.1, Section 203. Provide a car frame and platform of Manufacturers standard welded steel unit. Frame shall be constructed of structural of formed steel shapes welded or bolted together, consisting of cross head and bolster with channel uprights, gusset plates and diagonal bracing.

NOTE: State of Louisiana prefers roller guides on all Passenger Elevators. Freight elevators may be specified with guide shoes. Automatic Guide Rail Lubricators shall not be specified with roller guides. Omega rails should never be specified.

A. ROLLER GUIDES: Each guide shoe shall consist of rollers on a metal case. The rollers shall run on the three finished guide rail surfaces. Tires shall be neoprene or polyurethane.

B. "T" SHAPED CAR GUIDE RAILS: Provide new car guide rails. The rails shall be aligned with maximum deviation of 1/8 inch from plumb in all directions. Rail support brackets, fish plates, rail fastenings and bracket fastenings shall be provided.

Rails shall be thoroughly cleaned and shall be free of any signs of rust, grease or abrasion before final inspection. All guide rail joints shall be filed to assure perfectly matching surfaces. The base and shank of the "T" sections shall be painted with two coats of black enamel paint. Rails shall be of the length to extend from the bottom of the pit to the underside of the roof over the hoistway.

C.WORKMAN'S LIGHT AND OUTLETS: Provide duplex 3-wire grounded type receptacles, light switch and

lamp with wire guards on top of elevator car, beneath platform and the elevator pit area. Switches in car operating panel shall separately control light fixtures in car and lights on top and bottom of car. Pit light switch and receptacle shall be provided in accordance with CODE Rule 106.1e Receptacles shall be 2-pole, 3-wire grounded type rated for 15 amperes and 125 volts.

2.13 POWER UNIT, OIL CONTROL UNIT AND STORAGE TANK

2.13.1 POWER UNIT: (Oil Pumping and Control Mechanism) shall be compactly and neatly designed with all of the components listed below combined in a self-contained unit; oil reservoir with tank cover and controller compartment with cover; a submerged oil-hydraulic pump; an electric motor; an oil control unit with the following components built into a single housing; a high pressure relief valve; a check valve; an automatic unloading up start valve; a lowering and leveling valve; and a magnetic controller.

PUMP shall be especially designed and manufactured for oil-hydraulic elevator service. It shall be of the positive displacement type, inherently designed for steady discharge without pulsations to give smooth and quiet operation. Output of pump shall not vary more than 10% between no load and full load on the elevator car. Operating pressure shall not exceed 400 psi. Drive shall be by direct coupling. Provide a self-cleaning strainer in suction line to pump.

MOTOR shall be especially designed for oil-hydraulic elevator service and shall not exceed nameplate full load current by more than 10% and be continuously rated 120 starts per hour without exceeding The permissible temperature limitation, of standard manufacture, and of duty rating to comply with herein specified speeds and loads.

DRIVE shall be by direct coupling with the pump and motor submerged in the oil reservoir or by multiple V-belts and sheaves of number and size to insure maximum factor of safety. Drive type shall be determined based primarily on the load on the car, travel and speed.

CONTROLLER for the power unit shall contain all necessary electrical contractors, electrical mechanical switches, and thermal overload relays. All components shall be mounted in a NEMA 1 enclosure. Logic control system shall be microprocessor based, integrated solid state circuitry. The microprocessor-based system shall be protected from environmental extremes and excessive vibrations.

MOTOR STARTER: Elevator Contractor shall provide a solid state starter to limit current in-rush during starting and to provide gradual acceleration of the motor. Motor starting shall not be initiated by mechanical contacts. The starter shall include a current limit adjustment range of 200% to 450% of the overload adjustment range. An integral fault detection and diagnostic system shall be provided. It the internal fault detection system detects a failure, power shall be removed from the motor.

2.13.2 OIL CONTROL UNIT: Oil Control Unit shall consist of the following components, all built into a single housing. Welded manifolds with separate valves to accomplish each function will not be acceptable under this specification. All adjustments shall be accessible and shall be made without removing the assembly from the oil line. The Unit shall contain the following valve assemblies:

A. Furnish Relief Valve for hydraulic overload protection, which shall be externally adjustable, and shall bypass the oil flow without increasing back pressure by more than 10 percent above that required to open the valve. Bypass and relief valve shall be adjusted in accordance with ASME A17.1 Rule 303.2.
B. Up Start and Stop Valve shall be externally adjustable, and designed to bypass oil flow during start and stop of motor pump assembly. Valve shall close slowly, gradually diverting oil to or from the Jack Unit, insuring smooth up starts and up stops.
C. Safety Check Valve shall be designed to close quietly without permitting any perceptible reverse flow. It shall be designed to support the elevator with specified load at any point when pump stops or pressure drops below minimum operating levels on a positively locked column of oil when car is at rest.
D. Lowering Valve and Leveling Valve shall be externally adjustable for drop - away speed, lowering speed, leveling speed and stopping speed to insure smooth "DOWN" starts and stops. The leveling valve shall be designed to level the car to the floor in the direction the car is traveling when slowdown is initiated.
E. Manual Lowering Valve shall be capable of lowering the elevator car in event of power failure. The manual lowering valve shall be arranged to limit the maximum descending speed under manual operation to 25 fpm. The operating handle shall be marked in red.
F. Means shall be provided for isolating oil in power unit tank to insure ease of servicing and adjusting the elevator mechanism without removing the oil from the tank.

G. Service Gate Valves (oil line shut off valves) shall be installed in the oil supply line in the pit near the jack and in the machine room capable of withstanding 150 percent of design operating pressure. Gate valves shall have handles attached to the valves.
H. Provide ¼-inch pipe size valve quick release connection for attaching an inspector's pressure gauge.

2.13.3 STORAGE TANK: Storage Tank shall be constructed of steel, and shall be provided with a steel cover, protected vent opening, overflow connection and valve drain connection. Tank shall act as a storage tank only. Suitable gauge glass shall be provided if the top of the tank is over 4 feet above the floor. An initial supply of oil, sufficient for proper operation of the elevator shall be provided. The tank shall have a capacity equal to the volume of oil required to lift the elevator to top terminal plus a reserve of not less than 10 gallons. The permissible minimum liquid level shall be clearly indicated. The manufacturer's recommendation of the type of oil to be used shall be included in the written instructions for the care, adjustment and maintenance of the equipment. Self-cleaning strainers shall be provided to prevent foreign materials from lodging in control system. The viscosity of the hydraulic fluid shall be thermostatically controlled to maintain the fluid temperature in the reservoir, pump and valves at a constant operating viscosity.

2.13.4 SOUND REDUCTION: Sound-insulating panels, where pump and motor are not submerged, shall be provided and manufactured of reinforced 14-gauge steel with a 1" (25 mm) thick 1-1/2" core of fiberglass affixed to interior shall be mounted on all four open sides of the power unit frame.. A minimum of two sound isolating couplings shall be installed in the oil line in the machine room between the pump and the jack. Couplings shall be designed and manufactured in such a manner that they will be blowout proof.

Oil-hydraulic silencer (muffler type) shall be installed in the oil line near power unit and contain pulsation absorbing material surrounded by a blowout-proof, non-hammering, muffler housing arranged for inspecting interior parts without removing unit from oil line. Rubber hose without blowout proof features will not be acceptable. The power unit assembly shall be mounted on anti-vibration pads to isolate the unit from the building structure.

Mount rotating and vibrating equipment on vibration-isolating mounts designed to effectively prevent transmission of vibrations to structure and thereby eliminate sources of structure-borne noise from elevator system.

NOTE: State of Louisiana does not want a holeless jack application.

2.14 HYDRAULIC JACK, HOLE EXCAVATION AN CYLINDER PROTECTION

2.14.1 HYDRAULIC JACK UNIT: Jack unit shall be designed and constructed of sufficient size to lift the gross load to the height specified and shall be factory tested to insure adequate strength and freedom from leakage. The jack cylinder shall be accurately positioned and plumb with the guide rails. No brittle material, such as gray cast iron, shall be used in the jack construction. The jack unit shall consist of the following:

A. Telescoping plunger fabricated of heavy seamless steel tubing accurately turned and polished.
B. Stop ring electrically welded or screwed to the plunger to positively prevent plunger from leaving the cylinder.
C. Internal guide bearing.
D. Packing or seal of suitable design and quality.
E. Drip ring around the cylinder top.
F. Outer cylinder made of steel pipe and provided with a pipe connection and air bleeder.
G. Brackets welded to the jack cylinder for supporting the elevator on pit channels.
H. Provide an auxiliary safety bulkhead in the lower end of the cylinder.
I. Bleeder valve located at the cylinder on the jack unit to release air or other gases from system.
J. Elevator Contractor shall provide a Scavenger Pump with a copper tubing scavenger line electrically operated pump, installed between piston drip ring and oil storage tank. Scavenger pump shall be equipped with a water float designed to prevent operation of the pump should the pit flood. It shall also be designed to be manually reset. Pump and reservoir shall be strapped to the pit floor.

2.14.2 JACK CYLINDER PROTECTION: The jack cylinder and any underground piping shall be double wrapped with an approved applied extruded coating designed to help protect it from electrolytic and chemical corrosion.

The applied extruded coating shall include the following steps as a minimum:

A. Clean the bar pipe exterior surfaces;

B. Apply a minimum of 0.010 inch undercoating of heated butyl rubber adhesive;

C.. Apply a minimum of 0.040 inch thickness over coating of polyethylene, hot extruded over the undercoating:

D. The over coating shall be free of surface blemishes, cracks, voids, and contamination from foreign substances.

E. Pipe joints and coating repairs shall be sealed at the factory or on-site with coat with heat-shrinkable pipe sleeves as per cylinder manufacturers instructions;

F. Protect the applied extruded coating from damage until the cylinder is set into the cylinder well, plumbed and aligned.

2.14.3 CASING AND HOLE FOR JACK UNIT: Excavation for Jack: Drill excavation in the elevator pit to accommodate installation of cylinder; comply with applicable requirements in Division 2 Section "Earthwork." Provide a 0.25 inch steel well casing at least five inches larger in diameter than the PVC liner which will accommodate the jack cylinder. The well casing shall be accurately positioned and centered to within (01) one inch of the center of the car platform, plumbed and set so as to accept the PVC liner and jack cylinder. Area between the casing and cylinder shall be filled with washed dry sand after the cylinder has been accurately located. The bottom of the well casing shall be capped by continuous weld or plugging with a minimum of six inches of concrete embedded at the well casing bottom. The top of the casing shall be sealed. Elevator Contractor shall provide the jack hole complete, including drilling, pumping and anchorage of the completed assembly to the floor. Removal of excavation spoils deposited at the elevator pit will be the responsibility of the Elevator Contractor.

2.14.4 PVC LINER: Provide Schedule 80 PVC pipe liner with bottom cap and couplings; joints sealed watertight using PVC pipe manufacturer's recommended adhesive or heat welding methods. Liner inside diameter shall be not less than 3 inches larger than the elevator jack cylinder maximum outside diameter. Line bottom of cylinder well casing with minimum of six inches of fine, dry, salt-free sand. Set PVC liner into well casing, centered and plumb. Add twelve inches of fine, dry, salt- free sand between the steel well casing and the PVC liner to securely locate liner bottom end. Casing for Underground PVC piping complying with ASTM D 1785 joined with PVC fittings complying with ASTM D 2466 and solvent cement complying with ASTM D 2564.

2.14.5 CYLINDER INSTALLATION: Remove surface moisture from inside of the liner by wiping with a dry cloth or purging with warm dry air prior to installing elevator jack cylinder. Install Jack Cylinder. Provide elevator manufacturer's recommended supports under the cylinder head and attach cylinder head supports to cylinder and pipe structure in accordance3 with elevator manufacturer's instructions. Set cylinder and pit structure in accordance with elevator manufacturer's instructions. Set the jack cylinder plumb and level within one inch of center of the car platform.

2.14.6 LINER UPPER CAP: Provide PVC and cap for upper end of the PVC liner. Attach to the liner, using PVC cement or heat-welding method previously used for couplings and lower end cap. Pressure Test with air pressure and repair if necessary.

2.14.7 Corrosion Protective Filler: A solventless, petroleum-based gel formulated for filling the space between hydraulic cylinders and protective casings. Filler is heavier than water, electrically nonconductive, and liquefies at approximately 150 deg F.Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, the following:

Diversified Enterprises; No-Ox-Id R-R #6110A.
Pacific Standard Chemical Co.; Union-Gard 160.

Alternate Corrosion Protection Filler: Bidders may submit for an alternate filler material such as Union-Guard white powder type filler provided full cut sheets and product data is submitted. If approved, it will be noted in the next addendum.

2.14.8 SEAL: Seal gap between cylinder and PVC cap with black room temperature vulcanizing silicone sealant, forcing sealant into gap to full depth of cap wall thickness. Seal gap between casing and pit floor with plastic backer bead rod and one inch depth of silicone sealant, for gaps up to 0.75 inch in radius or cement grout full depth of pit floor for larger gaps. Seal gap between liner and well casing by same method specified of sealing gap between casing and pit floor.

2.14.9 GROUT: Non-shrink, Nonmetallic Grout: Factory-packaged, non-staining, non-corrosive, nongaseous grout complying with ASTM C 1107.

2.15 EMERGENCY OPERATION AND SIGNALING:

2.15.1 TWO-WAY COMMUNICATIONS: Emergency Communication System: Provide system that complies with ASME A17.1 and the U.S. Architectural & Transportation Barriers Compliance Board's "Americans with Disabilities Act (ADA), Accessibility Guidelines (ADAAG)," meeting the requirements of Southeastern Louisiana University telephone system. On activation, system buzzes the MPO and identifies elevator location. System provides two-way voice communication without using a handset and provides visible signals that indicate when system has been activated and when monitoring station has responded. System is to be contained in flush-mounted, recessed cabinet, with identification, instructions for use, and battery backup power supply. The entire communication assembly shall be Federal Communication Commission approved.

Fire Department Communication System: Provide telephone jack in car and required conductors in traveling cable for fire department communication system specified in Division 16 Sections.

2.15.2 FIRE SERVICE FOR ALL ELEVATORS: PHASE I & II OPERATION: Provide all elevator controls that shall meet all requirements of the latest edition of A17.1 Rule 211.3 for firefighters' service. Emergency operation to return the elevator to the designated primary landing at the 1st floor by means of a key operated switch in the main lobby. The designated alternate floor shall be the 2nd floor. Fixtures for the key switches to initiate Special Emergency Service shall be furnished.

2.16 TRAVELING CABLES:

A. Traveling cables from junction box on car to junction box in hoistway or directly to controller shall consist of flexible traveling cables conforming with requirements of NEC (NFPA 70). Junction boxes in hoistway and on car shall by equipped with terminal blocks. All connections to terminal blocks shall be made with either terminal eyelet connections or pressure wire connectors of the clamp type that meet UL 486A requirements for stranded wire. Terminal blocks shall have permanent indelible identifying numbers for each connection. The outer covering must remain intact between junction boxes. Abrupt bending or twisting producing distortion of cable shall not be allowed. Cables shall be free from any possible contact with hoistway structure, car, or other equipment. Where necessary, shields or pads shall be provided to protect the cables.
B. Provide 10 percent, but not less than 2, spare conductors in each traveling cable. Tag spares so they can be identified.
C. Provide separate traveling cables for car lighting and control circuits.
D. Provide two sets of shielded coaxial conductors in the traveling cable in addition to those wires specified for the two-way communication device in the elevator car or supply a separate cable for the two-way communication device. Cable shall extend from junction box in hoistway or controller to two-way communication device in the car.
E. Conductors for alarm and emergency stop shall not be in the same traveling cable.
F. Provide two sets of conductors for car lighting and ventilation in separate traveling cables.

2.17 CONTROL, COMMUNICATION AND SIGNAL SYSTEM IDENTIFICATION:

A. Install a permanent wire marker on each wire at each termination.
B. Identifying numbers and letters on the wire markers shall correspond to those on the wiring diagrams used for installing the systems.
C. Wire markers shall retain their markings after cleaning.

2.18 BUFFERS: Buffers shall be of the spring type, of a design suitable for the depth of the pit. They shall be mounted on continuous channels fastened to the elevator guide rail or securely anchored to the pit floor and substantial extensions will be provided, if required. Buffer anchorage at pit floor shall be arranged to avoid puncturing of the pit waterproofing.

2.19 PIT STOP SWITCH: ASME A17.1 RULE 106.1f: Provide an enclosed stop toggle switch in the pit of each elevator that is readily accessible from access door to pit. Locate adjacent to pit access ladder at top rung. Toggle switch shall be manually opened and closed with red operating handle conspicuously and permanently marked "STOP" and "RUN". Switch shall be positively opened mechanically, and opening shall not be solely dependent on springs.

2.20 PIT LADDER: ASME A17.1 RULE 106.1f To be installed by others. Provide a safe and convenient access to each elevator pit that shall conform to the following:

A. Provide a fixed vertical ladder of noncombustible material, located within reach of the access door. Ladder shall extend not less than 42 inches above the sill of access door.
B. The rungs shall be at least 12 inches wide.
C. Provide a clear distance of not less than 4 2 inches from the centerline of the rungs to the nearest permanent object in back of the ladder.

2.21 KEYS: Elevator Contractor shall provide to PROJECT ADMINISTRATOR (05) five sets of all keys pertaining to the operation of the elevators. The keys shall include firefighter service, independent service, stop switch, inspection, hoistway access, fan, lighting, and any other key for the elevators and control panels.

ART 3. ELECTRICAL

3.1 ELEVATOR WIRING AND ELECTRICAL INSTALLATION

3.1.1 ELEVATOR WIRING: Wring shall be provided for all electrically operated items of elevator equipment and shall comply with the requirements of NFPA 70. The size wire permitted shall be No. 18 AWG for control and signal circuits and No. 12 AWG for power and lighting circuits. Work light fixtures and grounded duplex receptacles shall be provided on the top and bottom of each car for 150-watt incandescent lamps. Fixtures shall have wire lamp guards and toggle switches. Work light fixtures and traveling cable junction boxes shall be so located that the work lights will provide illumination at the junction boxes.

3.1.2 NEW WIRING: Except for short sections, wiring, excluding traveling cables, shall be in rigid conduit, EMT or duct. Short section of wire, which may require shifting to connect to switches and equipment may be installed inflexible conduit. Conduit or EMT shall terminate in junction boxes. Wires shall be identified and match symbols shown on the wiring diagrams. Control and signal wires shall be brought to accessible numbered terminal blocks on the controller. Intra-panel wiring shall be flame- resistant type.

3.1.3 **ELECTRICAL INSTALLATION FOR THE GENERAL CONTRACTOR TO SUPPLY FOR THE ELEVATOR CONTRACTOR.** Work described in this Section includes providing all labor, materials and quipment indicated, specified and necessary for a complete and operating electrical system nd related systems NOT INCLUDED UNDER THE ELEVATOR CONTRACTOR'S WORK.

A. Install systems in accordance with the manufacturer's recommendations, A17.1, A17.2, and NFPA 70.
B. Provide from electrical source suitable connections from the power mains to the controller, signal equipment feeders as required, including necessary circuit breakers, shunt trip and fused lockable main line disconnect switches.
C. Provide wiring to the controller for car lighting and ventilation.
D. Heat and smoke sensing devices at elevator lobbies on each floor with electrical conductors terminating at a properly marked panel in the elevator panel in the elevator machine room.
E. Telephone connection to the elevator controller.
F. Wiring from elevator hoistway to outside alarm bell as required by the Safety Code for Elevators and Escalators (ASME A17.1).

ART 4. INSTRUCTIONS: CONTRACTOR shall furnish Maintenance and Operating manuals, verbal instructions, Preventive Maintenance and Inspection Schedules, Special Tools, Safety Considerations, Repair Requirements, Parts List, Spare Parts Data, and As-built Drawings for each piece of equipment furnished to the ENGINEER.

4.1.1 MANUALS: Maintenance and operating manuals, three (3) copies each, for each separate piece of equipment hall be delivered to the ENGINEER not less than 20 days prior to final inspection. Manuals shall e complete detailed guides for the maintenance and operation of the equipment. They shall nclude complete information necessary for check-out, starting, repair and replace components f the systems, adjusting, trouble shooting, diagnostic techniques, maintaining in continuous operation for

ng periods of time, dismantling and reassembling of the complete unit and sub-assembly components. Manuals shall lude an index covering all component parts clearly cross-referenced to diagrams and strations. Illustrations shall include "exploded" views showing and identifying each separate n. Emphasis shall be placed on the use of special tools and instruments. The function of each ce of equipment and the reason for each precaution shall be clearly set forth. Manuals must reference the exact odel, style and size of the piece of equipment and system being furnished. Manuals referencing equipment similar to but different model, style, and size than that furnished will not be accepted.

4.1.2 INSTRUCTIONS: CONTRACTOR shall provide qualified, factory-trained manufacturer's representatives to give tailed instructions to assigned personnel in the operation and complete maintenance for each piece of uipment. All such training shall be at the job site. Certain requirements are more specifically tailed in various technical sections.

Where specific details are not given, instruction periods, items shall be from two (2) to four (4) hours instruction for every piece of component equipment in a system. This is to assure proper instruction in the operation of errelated system. Twenty (20) calendar days prior to anticipated final inspection, ONTRACTOR shall submit a complete proposed training schedule to the ENGINEER for coordination.

All instruction periods shall be considered concluded only when the ENGINEER is satisfied in regard to complete nd thorough coverage. No training shall be actually accepted for an item of equipment until (1) e manual for the equipment has been in the hands of the ENGINEER for at least two weeks and) the equipment if fully operational and has been tested to the satisfaction of ENGINEER. All aining unless specifically noted otherwise, shall be completed prior to acceptance of the project. he ENGINEER reserves the right to request the removal of, and substitution for, any instructor whom, in the opinion of the ENGINEER, does not demonstrate sufficient qualifications in accordance with requirements for struction above.

4.1.3 PREVENTATIVE MAINTENANCE AND INSPECTION SCHEDULE: Procedures and instructions ertaining to frequency for preventive maintenance, inspection, adjustment, lubrication, nd cleaning necessary to minimize corrective maintenance and repair shall be turned over the ENGINEER before final acceptance as follows:

A. PREVENTATIVE Maintenance Plan and Schedule: Include manufacturer's schedule for routine preventive maintenance, inspections, tests, and adjustments required to ensure proper and economical operation and to minimize corrective maintenance and repair. Provide manufacturer's projection of preventive maintenance man-hours on a daily, weekly, monthly, and annual basis.
B. SPECIAL TOOLS: List of special tools and test equipment required for maintenance and testing of the products supplied by the CONTRACTOR.
C. LUBRICATION DATA:
   1. Include lubrication data table showing recommended lubricants for specific temperature ranges and applications.
   2. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities; and
   3. Lubrication schedule showing service interval frequency.
D. CORRECTIVE MAINTENANCE: Include manufacturer's recommendations on procedures and instructions for correcting problems and making repairs.
E. TROUBLE SHOOTING GUIDE AND DIAGNOSTIC TECHNIQUES: Include step-by-step procedures to promptly isolate cause of typical malfunctions. Describe clearly why checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment may be re used or require replacement.
F. WIRING DIAGRAMS AND CONTROL DIAGRAMS shall be point-to-point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams number electrical and electronic wiring and the terminals for each type, identically to actual installation numbers.
G. MAINTENANCE AND REPAIR PROCEDURES: Include instruction and list tools required to restore product or equipment to proper condition or operating standards.
H. REMOVAL OR REPLACEMENT INSTRUCTIONS: Include step-by-step procedures and list required tools and supplies for removal, replacement, disassembly of components, assemblies, sub assemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Instructions shall include a combination of text and illustrations.
I. SPARE PARTS AND SUPPLY LISTS: Include lists of spare parts and supplies required for

maintenance and repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead time to obtain.

J. SAFETY CONSIDERATIONS: Information relating to load limits, speeds of operation, environmental criteria (temperature and pressure limitations), and personnel hazards and equipment safety precautions.

K. REPAIR REQUIREMENTS: Instructions necessary to check-out, trouble shoot, repair, and replace components of the systems, including integrated electrical and mechanical schematics and diagrams an diagnostic techniques, controller and drive fault interpretations and codes necessary to enable operation and trouble shooting after acceptance of the system.

L. PARTS LIST: List of parts and components for the system by manufacturer's name, part number, nomenclature, and stock level required for maintenance and repair necessary to ensure continued operation with minimal delay.

M. IDENTIFICATION AND DATA RETRIEVAL: The Elevator Contractor shall have an identification and data retrieval system. Records, Drawings, Submittals, and Equipment shall be identified to reference the following:

1. Contract number
2. Contract specification number
3. Contract drawing number
4. Submittal document number
5. Contract change number
6. Contractor's drawing number system

N. AS BUILT DRAWINGS that provide current factual information including deviations from, and amendments to the drawings and changes in the work, concealed and visible shall be submitted to the ENGINEER

ART 5. FINAL INSPECTIONS AND TESTS

5.1 FINAL INSPECTION AND TESTS: Final inspections and tests shall be performed as follows:

A. TEST NOTIFICATION: When the elevator work included in the contract if fully complete. Prior to scheduling final inspections and tests, a minimum 72-hour written notification shall be given by CONTRACTOR to ENGINEER, that elevators are ready for final inspection and tests. ENGINEER will notify Mr. Chuck Johnson, State of Louisiana, Office of State Buildings, phone number (225) 219-4783, fax number (225) 219-4810, to schedule a Certified Elevator Inspector under contract with the State to coordinate inspections and test with CONTRACTOR and ENGINEER. CONTRACTOR shall be fully responsible to perform all tests and demonstrate the proper operation of all parts and provisions of the equipment.CONTRACTOR shall prove to the satisfaction of ENGINEER and the Elevator Inspector that the elevator, as installed, complies with the requirements of this contract and all applicable requirements of ASME A17.1

B. INSPECTION PROCEDURE: Perform acceptance inspections and tests in accordance with ASME A17.1, Section 1006. The inspection procedure outlined in ANSI A17.2 shall form a part of the final inspection. CONTRACTOR shall demonstrate that the performance of the elevator as specified has been provided.

C. TESTING MATERIALS AND INSTRUMENTS: The CONTRACTOR shall furnish all test instruments, gauges, and material required for final inspection. Include standard 50-pound test weights, and insulation "Megger" 600 volt, alternating current volt meter and ammeter, three (03) Celsius calibrated thermometers, spirit level, stop watch, and a direct reading tachometer.

D. DIAGNOSTIC TESTING DEVICE: A diagnostic testing device or maintenance terminal suitable for all trouble shooting procedures related to the specific type microprocessor controls installed on this project, shall be provided. This diagnostic testing device or maintenance terminal shall be demonstrated and tested during final testing of the elevator installation. A series of not less than ten simulated malfunctions shall be diagnosed properly by the device. A period of at least 2 hours shall be dedicated to the instruction of its use to ENGINEER

1. After successful testing to the diagnostic device in conduction with the microprocessor controls, the testing device shall become the property of the Southeastern Louisiana University.
2. CONTRACTOR shall provide Password or Code to access control equipment to ENGINEER.

E. FINAL INSPECTION: In addition to any other tests, make the following tests at the time of the final inspection, as follows:

1. TEST PERIOD: Subject elevator to a test for a period of one-hour continuous run, with specified rated load in the car. During the test run, the car shall be stopped at all floors in both directions of travel for a standing period of 10 seconds per floor. Provide a manual test of the final limits (up and down over travel). No component will be permitted to fail.
2. SPEED LOAD TEST: Determine the actual speed of the elevator car, in both directions of travel with the rated load and with no load in the elevator car. Make speed tests before the rated-load test run and also after the rated load test run. Determine speed by applying a tachometer. The actual measured speed of the elevator car with the rated load in the up and down direction shall be within 5 percent of the rated contract speed. The maximum difference in actual measured speeds obtained under the various conditions outlined shall not exceed 5 percent of the total difference between the "UP" and "DOWN" directions.
3. CAR LEVELING TESTS: Test elevator car leveling devices for accuracy of landing at al floors with no load in the car, symmetrical load in the car, and with the rated load in the car in both directions of travel. Determine accuracy of floor landing both before and after the rated full-load run test. Measure leveling tolerances, using a 1800 mm straightedge laid flat on higher of two surfaces, shall be plus or minus 1/8 inch under ever rated loading condition.
4. INSULATION RESISTANCE TESTS: Complete-wiring systems of the elevator shall be free from short circuits and grounds. The insulation resistance shall be determined by use of a "Megger". Conductors shall have an insulation resistance of not less than one megohm between each conductor and all other conductors.
5. TEMPERATURE RISE TESTS: Determine temperature rise of machine enclosure, pump motor and hydraulic oil in the storage tank, during the full-load test run for a minimum of one hour. Under these conditions the temperature rise of the equipment shall not exceed 55 degrees C. above ambient temperature. Start the test only when all parts of equipment are within 5 degrees. C of ambient temperature, at the time of starting the tests.
6. REPORTS: Test reports on the following tests shall be submitted for the elevator systems in accordance with ASME A17.1 and A17.2.
   a. Continuous - operations tests
   b. Temperature rise test
   c. Acceptance Inspection and Test
7. RECORDS: The CONTRACTOR shall provide on-site records of each inspection and test performed throughout the life of the contract.

CONTRACTOR shall keep records that include, but no limited to, factual evidence that the required inspections and/or tests have been performed, including type and number of inspections or tests involved, identification of operators and inspectors, result of inspections or tests, nature of defects, causes for rejection, proposed remedial action, and corrective actions taken.

5.1.2 CERTIFICATION: Perform and furnish certification for the tests required by A17.1 Section X and Rule 1006. Certification must state that each car safety device and relief valve had been tested and approved for use with the equipment having the specific ratings indicated or specified.

5.1.3 RE INSPECTION: If any equipment is found to be damaged or defective, or if the performance of the elevator does not conform with the requirements of the contract specifications or the Elevator Safety Code, no certificate of approval shall be issued, until all defects have been corrected. When non-conforming items, parts, test failures, defects, repairs and adjustments have been completed and the discrepancies corrected, the ENGINEER shall be notified and the elevator will be re inspected at the Elevator Contractor's expense. Do not use a rejected elevator until it has been re inspected and approved.

END